



The Dynamic Sulfide Saturation Process and a Possible Slab Break-off Model for the Giant Xiarihamu Magmatic Nickel Ore Deposit in the East Kunlun Orogenic Belt, Northern Qinghai-Tibet Plateau, China

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Abstract

The Xiarihamu Ni deposit is located in the East Kunlun orogenic belt in the northern part of the Qinghai-Tibet plateau. The regional tectonic setting, lithology, and mineralogy suggest that the Xiarihamu Ni deposit formed in a postcollisional setting. We use a slab break-off model to explain the simultaneous appearance of the Xiarihamu magmatic Ni sulfide deposit and the high-pressure eclogite exhumation in the East Kunlun orogenic belt. In situ $\delta^{34}\text{S}_{\text{V-CDT}}$ value of pyrite from the country rock (granitic gneiss) of the Xiarihamu complex is 11.2‰, whereas in situ $\delta^{34}\text{S}_{\text{V-CDT}}$ value of sulfides from the Xiarihamu I complex range from 2.4 to 7.7‰, with an average of 4.5‰. Crustal sulfur most likely accounts for approximately 40 to 60% of the total sulfur content in the Xiarihamu magma. Our modeling shows that the Xiarihamu parental magma may have achieved sulfide saturation and experienced approximately 0.005% sulfide segregation before complex emplacement. Based on previous studies and the platinum group element data in this study, we believe that the Xiarihamu magma was sulfide saturated when olivine began to crystallize. The zones with rapidly declining olivine Fo and clinopyroxene Mg# values correspond well with the locations of orebodies, implying that the crystallization of olivine or clinopyroxene played a role in the sulfide saturation. The whole-rock Ni content of the peridotite is inversely related to the Cr# and FeO contents in spinel, and the whole-rock Ni content of the websterite is negatively correlated with orthopyroxene Mg#; thus, during the crystallization of spinel and orthopyroxene, more sulfide formed. Our calculation shows that the decrease in sulfur content at sulfide saturation (SCSS) during fractional crystallization depends mainly on a decrease in temperature, and the change in the melt composition is a secondary or minor factor. Although the contribution of fractional crystallization to total sulfur content may be only approximately 3.3 to 6.0%, an increasing concentration of sulfides were accumulated at the latest crystallization in the host rock because of the incompatibility of sulfur. Recognizing this phenomenon is important for ore exploration. We logged 11 drill cores meter by meter and prepared 110 thin sections from across these 11 drill cores. The lithofacies distribution and crosscutting phenomena have been clearly determined. Combined with systematic geochemical data on the cross section, this paper summarizes the mineral crystallization sequence in different locations of the Xiarihamu I mafic-ultramafic magma chamber and depicts a six-stage model for sulfide saturation during the consolidation process of the Xiarihamu complex.

Introduction

The giant Xiarihamu magmatic sulfide Ni ore deposit, which occurs in the northern part of the Qinghai-Tibet plateau (Fig. 1a), contains ~157 million metric tons (Mt) of sulfide ores with average grades of 0.65 wt % Ni, 0.013 wt % Co, and 0.14 wt % Cu (No. 5 Geological and Mineral Survey Institute of Qinghai Province, 2014). Thus, the Xiarihamu deposit is not only the second largest Ni deposit in China but also one of the 20 largest known magmatic sulfide deposits in the world (Jiang et al., 2015; Li et al., 2015; Song et al., 2016; Zhang et al., 2017b).

Large amounts of mafic-ultramafic magma produced by high degrees of partial melting of the mantle, probably associated with the impact of mantle plumes on the lithosphere, are believed to be necessary for the formation of magmatic sulfide deposits, such as the Noril'sk Ni-Cu-PGE (platinum-group elements) deposit and the Pechenga Ni-Cu-PGE deposit (Russia), the Voisey's Bay Ni-Cu-Co deposit (Canada), the

Jinchuan Ni-Cu deposit (China) (Walker et al., 1997; Zhou et al., 2002; Pirajno, 2004; Li et al., 2005; Lightfoot and Keays, 2005; Saunders et al., 2005; Lightfoot and Milkereit, 2007; Hanski et al., 2011; Song et al., 2011; Liu et al., 2016). In contrast, the Ni-Cu deposits in orogenic belts, such as the Cu-Ni deposits in the eastern Tianshan (e.g., Huangshandong, Tulaergen, Huangshanxi (J.W. Mao et al., 2002; Zhou et al., 2004; Z.H. Zhang et al., 2008; Qian et al., 2009; M.J. Zhang et al., 2011; Gao and Zhou, 2013; Sun et al., 2013; Y.J. Mao et al., 2015), the Hongqiling Cu-Ni deposit (Wei et al., 2013), the Aguablanca deposit in the Variscan collisional orogeny, Spain (Casquet et al., 2001; Piña et al., 2013), the deposits in the Selebi-Phikwe belt, Botswana (Maier et al., 2008), are on average smaller in size and lower in metal grade than those occurring in large igneous provinces (Song et al., 2016; Begg et al., 2017). However, the Xiarihamu deposit is an exception in such a tectonic setting because it is a giant Ni deposit with a high average Ni metal grade of 0.65%. The paucity of new discoveries may reflect the unique confluence of geologic conditions required to concentrate sulfides in this environment.

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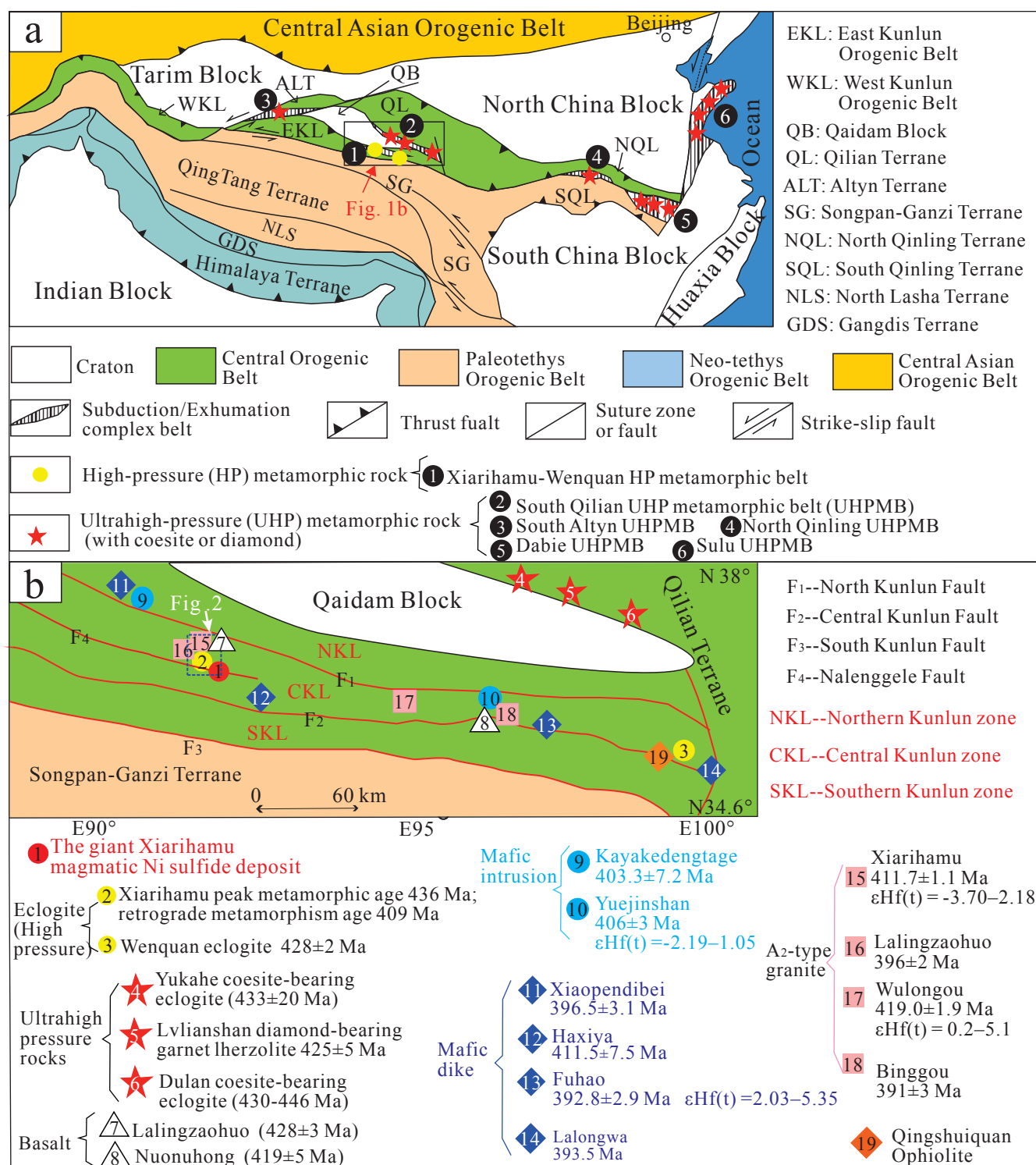


Fig. 1. a. Tectonic sketch map of China (modified after Yang et al., 2011). b. Simplified tectonic units of the East Kunlun orogenic belt (modified after Xu et al., 2001). Age and Hf isotope resources: Xiarihamu eclogite (Zhang et al., 2017a); Wenquan eclogite (Meng et al., 2013); Yukahe coesite-bearing eclogite, UHP metamorphic age 433 ± 20 Ma (Song et al., 2010); Lvlianshan diamond-bearing garnet lherzolite, UHP metamorphic age 425 ± 5 Ma (Song et al., 2005); Dulan coesite-bearing eclogite, UHP metamorphic age 430 to 446 Ma (Zhang et al., 2010); Nuomuhong basalt (Zhu et al., 2006); Lalingzaohuo basalt (Qinghai Geological Survey, 2012); Kayakedengtage mafic intrusion (Chen et al., 2006); Yuejinshan mafic intrusion (Liu et al., 2012); Xiaopendibei mafic dike (Zhang et al., 2013); Haxiya mafic dike (Yang et al., 2014); Fuhao mafic dike (Xiong et al., 2014); Lalongwa mafic dike (Sun et al., 2004); Xiarihamu A2-type granite (Wang et al., 2013; Gan, 2014); Lalingzaohuo A2-type granite (Chen et al., 2013); Wulongou A2-type granite (Yan et al., 2016); Binggou A2-type granite (Liu et al., 2013b).

A unique phenomenon for the Xiarihamu Ni deposit is that eclogite is present near the Xiarihamu Ni deposit. The eclogite exhibits a 438 Ma high-pressure eclogite facies metamorphic peak age during continental collisions and a 409 Ma retrograde metamorphism age during the continental crust exhumation process (Zhang et al., 2017a). Interestingly, the ore-bearing websterite in the Xiarihamu deposit formed at 411.6 ± 2.4 or 406.1 ± 2.7 Ma (Li et al., 2015; Song et al., 2016). Is the simultaneous appearance of the Xiarihamu magmatic Ni sulfide deposit and the eclogite exhumation in the same tectonic belt a coincidence? What is the relationship between deep subduction/exhumation and the formation of the Xiarihamu Ni deposit? This paper attempts to answer these questions.

Fully understanding the sulfide saturation process of a magmatic system requires a clear recognition of the relationship between the lithofacies distributions and the orebodies. The works of Leshner (1989) and Lightfoot et al. (2012) provide good examples. The lithofacies distribution section, such as Li et al. (2015, fig. 4), Zhang et al. (2017b, fig. 2), and Song et al. (2016, fig. 4), is based on the original section figures from the No. 5 Geological and Mineral Survey Institute. Based on the detailed field records of 11 drill cores compiled meter by meter in a typical section and the observation of 110 thin sections, this paper presents a new understanding of the relationship between the host rocks and different orebodies, which differs from what was previously believed. In addition, prior researchers did not pay much attention to the relationship between the geochemical stratigraphic variations and the lithofacies distributions. This paper systematically analyzed the major and minor element compositions of the minerals, the in situ S isotope values of the sulfides, the Sr isotope values, and the PGE compositions of whole-rock samples from two representative drill cores in one cross section. The spatial variations in the geochemical compositions of different lithofacies and orebodies have been determined. In addition, the degree of crustal sulfur contamination and the temperature decrease and composition change during fractional crystallization on sulfide saturation are clearly discussed. Finally, this paper proposes a dynamic sulfide saturation process for the Xiarihamu I mafic-ultramafic complex.

Geologic Background

The Xiarihamu region is located in the East Kunlun orogenic belt in the northern part of the Qinghai-Tibet Plateau (Jiang et al., 2015; Li et al., 2015). The East Kunlun orogenic belt is located in the western part of the Central orogenic belt in mainland China (Fig. 1a). In the Central orogenic belt, five coesite- or diamond-bearing UHP metamorphic belts have been discovered (Fig. 1a): Dabie (Xu, 1987; Wang et al., 1989), Sulu (Enami, 1990), North Qaidam (Yang et al., 2002), South Altyn (Liu et al., 2002), and North Qinling (Yang et al., 2003). These UHP metamorphic belts have been interpreted to have been formed by the northward subduction of one continental block beneath the other (Zheng et al., 2012). For example, the Dabie-Sulu UHP metamorphic belt is considered to have been built by northward subduction of the South China block beneath the North China block in the Triassic, while the Altyn and North Qaidam UHP terranes were produced in the early Paleozoic by northward subduction of the Qaidam block

beneath the Tarim block and the Qilian terrane, respectively (Zheng et al., 2012). The North Qaidam UHP belt has been interpreted to have been exhumed by a slab break-off combined with a process of oblique extrusion during the transformation from normal to oblique intracontinental subduction between the Qilian and Qaidam terranes (Xu et al., 2006). The exhumation has been interpreted to have begun at 470 to 460 Ma and was completed by 406 to 400 Ma. Exhumation structures are well preserved in the UHP rocks and record extensive retrograde metamorphism (Xu et al., 2006).

The East Kunlun orogenic belt is bordered by the Qaidam block to the north, the Sonpan-Ganzi terrane to the south, and the Qilian terrane to the east (Fig. 1b). The North Qaidam ultrahigh-pressure (UHP) metamorphic belt is located at the border between the Qaidam block and the Qilian terrane (Fig. 1b). The Yukahe coesite-bearing eclogite with an UHP metamorphic age of 433 ± 20 Ma (Song et al., 2010), the Lvlianshan diamond-bearing garnet lherzolite with an UHP metamorphic age of 425 ± 5 Ma (Song et al., 2005), and the Dulan coesite-bearing eclogite with an UHP metamorphic age of 430 to 446 Ma (Zhang et al., 2010) compose the North Qaidam UHP metamorphic belt (Fig. 1b). UHP metamorphic rocks exposed at the surface in subduction zones and orogenic belts are evidence that crustal materials are tectonically transported to and exhumed from great depths, in some cases exceeding ~150 km (Erdman and Lee, 2014).

The East Kunlun orogenic belt trends east-west and is approximately 1,500 km long and 50 to 200 km wide. It is cut by three closely related, deep E-W-trending regional faults (F₁: the North Kunlun, F₂: Central Kunlun, and F₃: South Kunlun faults, from north to south, respectively; Fig. 1b; Huang et al., 1984; Jiang et al., 1992). The East Kunlun orogenic belt is further divided into the Southern, Central, and Northern Kunlun zones by the North Kunlun fault (F₁) and the Central Kunlun fault (F₂) (Fig. 1b).

The Xiarihamu Ni deposit is located in the Central Kunlun zone (Fig. 1b). The exposed rocks in the Central Kunlun zone include (1) granitic gneiss, marble, skarn and schist of the Paleoproterozoic Baishahe Formation; (2) Neoproterozoic granitic gneiss and schist; (3) sandstone, siliceous rocks, basalt, and dolomite of the Upper Silurian to Lower Devonian Qimantage Group; (4) conglomerate, siltstone, and rhyolite tuff of the Upper Devonian Maoniushan Formation; (5) sandstone and dolomitic limestone of the Lower Carboniferous Dagangou Formation; and (6) Cenozoic sediments (Fig. 2). In the Xiarihamu area, Neoproterozoic granitic gneiss has been discriminated from Paleoproterozoic gneiss via zircon U-Pb dating in recent years (Gan, 2014; Wang et al., 2016). The Neoproterozoic granitic gneiss is in close contact with the Xiarihamu I mafic-ultramafic complex (Fig. 2). The Upper Silurian to Lower Devonian Qimantage Group is further divided into the Lower, Middle, and Upper Formations (China University of Geosciences, Wuhan, 2011; Qinghai Geological Survey, 2012). The Lower Formation is composed of sandstone and siliceous rocks, representing bathyal-shallow marine environments; the Middle Formation consists mainly of basalt with a zircon U-Pb age of 428 ± 3 Ma (Fig. 2), where this basalt has been interpreted to be tholeiite in an extensional arc-back basin environment (China University of Geosciences, Wuhan, 2011; Qinghai Geological Survey, 2012); and the Upper

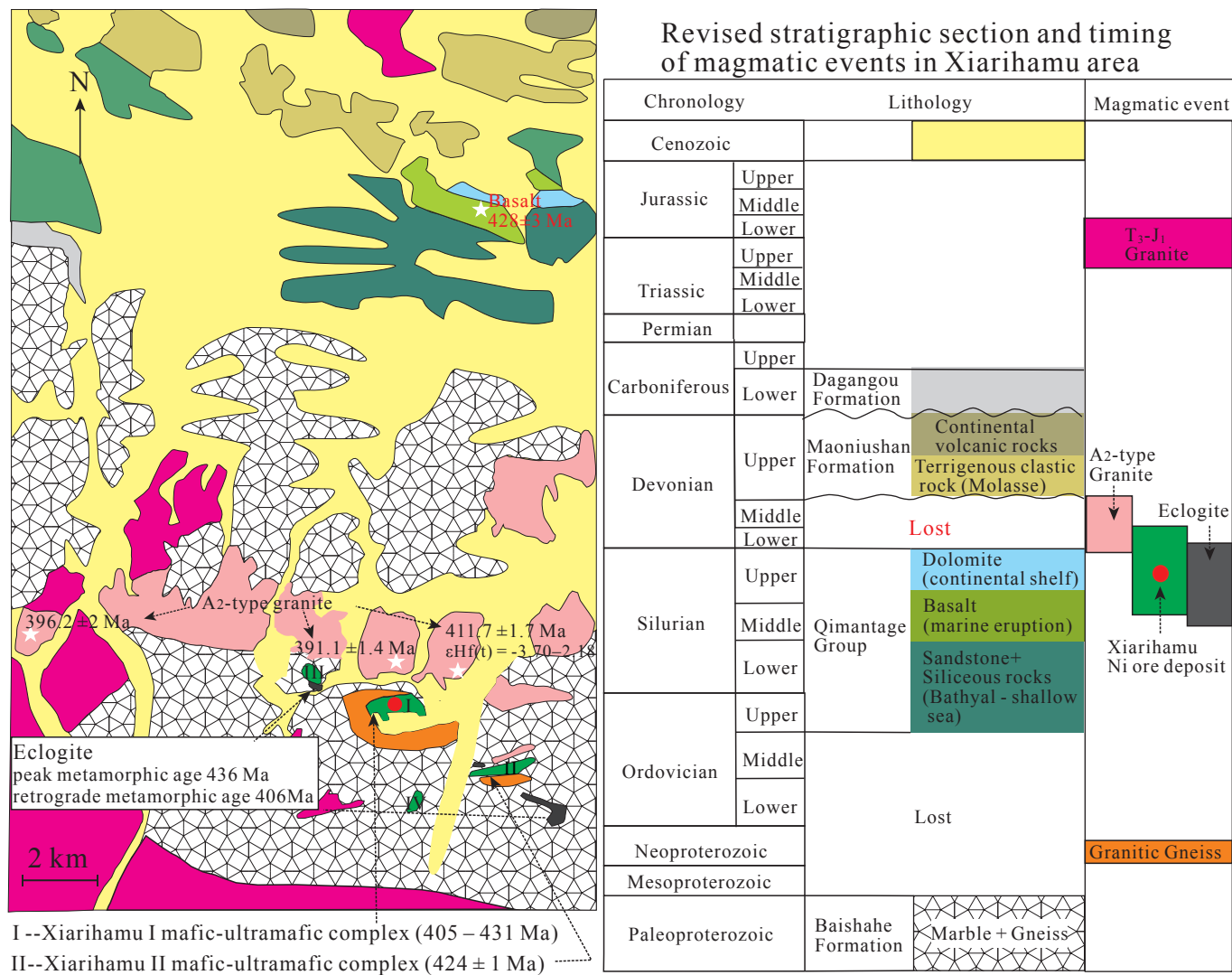


Fig. 2. Regional 1:50,000 geologic map of the Xiarihamu area (modified after China University of Geosciences, 2011, and Qinghai Geological Survey, 2012). Age and Hf isotope resources: the age of the Xiarihamu I mafic-ultramafic complex (Li et al., 2015; Song et al., 2016); the age of the Xiarihamu I mafic-ultramafic complex (Peng et al., 2016); Xiarihamu eclogite (Zhang et al., 2017a); Xiarihamu A₂-type granite (Chen et al., 2013; Wang et al., 2013; Gan, 2014); 428 Ma basalt (Qinghai Geological Survey, 2012); Neoproterozoic granitic gneiss (Gan, 2014; Wang et al., 2016).

Formation comprises dolomite and dolomitic limestone, representing a continental shelf environment (Fig. 2). The age of the back-arc tholeiitic basalt is very close to the age of the older gabbro (431 Ma; Li et al., 2015) in the Xiarihamu complex. No sedimentary strata from the Early and Middle Devonian are observed in our research area. The Upper Devonian Maoniushan Formation has two members: a Lower clastic member and an Upper volcanic member (Fig. 2). The Lower clastic member is composed of fuchsia-colored medium-grained quartz sandstone and coarse-grained feldspar-quartz conglomerate, representing lake-phase molasse deposition. The Upper volcanic member is composed of brecciated tuff and lava, suggesting a continental volcanic environment (Qinghai Geological Survey, 2012; Fig. 2). The Qimantage Group and the Maoniushan Formation are observed to the north of the Xiarihamu Ni deposit (Qinghai Geological Survey (QGS), 2012). The Qimantage Group is regarded as arc-back basin

strata (Ren, 2010; Wang et al., 2010; Gao et al., 2011), whereas the Maoniushan Formation has been interpreted to mark the end of an orogenic cycle and the beginning of the late Paleozoic basin phase (Mo et al., 2007; Li et al., 2008).

The Central Kunlun fault represents the remnant margin of a small early Paleozoic oceanic basin, and the youngest age of the Qingshuiquan ophiolite complexes in Central Kunlun is 508 Ma (Li et al., 2006). Furthermore, an arc-related granite in the East Kunlun region has been dated to 508 Ma (China University of Geosciences, Wuhan, 2003). Therefore, Mo et al. (2007) inferred that the East Kunlun region entered the oceanic subduction stage after 508 Ma. The transition from oceanic subduction to continent-continent collision was believed to have occurred at 438 Ma (Liu et al., 2013a). An eclogite with a peak metamorphic age of 428 Ma in the Wenquan area in the eastern part of this belt (Fig. 1b) and an eclogite with a peak metamorphic age of 436 Ma in the Xiarihamu deposit

have been interpreted to be associated with this continent-continent collisional event (Fig. 2; Meng et al., 2013; Zhang et al., 2017a). The Xiarihamu eclogite revealed a retrograde metamorphism aged 409 Ma, representing the exhumation age (Zhang et al., 2017a). Some nonmineralized mafic intrusions or mafic dikes (403–406 Ma; Fig. 1b), such as the Yuejingshan and Katakedengtage mafic intrusions (Chen et al., 2006; Liu et al., 2012), and some mafic dikes (411.5–382.8 Ma), such as Xiaopendibei and Haxiya (Sun et al., 2004; Zhang et al., 2013; Xiong et al., 2014; Yang et al., 2014), also exist near the northern boundary of the Central Kunlun zone (Fig. 1b). The Upper Silurian–Middle Devonian (411.7–391.1 Ma) A₂-type granite forms a belt near the Xiarihamu Ni deposit (Chen et al., 2013; Liu et al., 2013b; Wang et al., 2013; Gan, 2014; Yan et al., 2016; Figs. 1b, 2).

Geology of the Xiarihamu Ni Deposit

Lithofacies distribution and orebody locations

The Xiarihamu area contains four mafic-ultramafic complexes, including Xiarihamu I, Xiarihamu II, Xiarihamu III,

and Xiarihamu IV (Fig. 2). The Xiarihamu magmatic sulfide deposit is hosted within the Xiarihamu I complex (Fig. 2). The mineralization potential of the other three mafic-ultramafic complexes remains under investigation (Fig. 2; Peng et al., 2016), and the present study mainly focuses on the genesis of the Xiarihamu I mafic-ultramafic complex. The Xiarihamu I mafic-ultramafic complex was emplaced into Neoproterozoic granitic gneiss and schist (Figs. 2, 3). As previously noted, the Neoproterozoic granitic gneiss has been discriminated from the Paleoproterozoic gneiss in the Baishahe Formation via zircon U–Pb dating in recent years (Gan, 2014; Wang et al., 2016). The Xiarihamu I mafic-ultramafic complex is irregularly shaped, with a length of ~1,400 m, a width of ~900 m, and a depth of 300 to 600 m (Figs. 3, 4). The western part of the complex is not exposed, and the southern part is covered with Quaternary clastic sediments. The surface exposures of the complex are composed of harzburgite in the middle part, orthopyroxenite and websterite in the eastern parts, and gabbro in the north, as shown in Figure 3. Oxidized ore, such as the iron cap, was found in the websterite in prospecting lines 2 (L2) to 5 (L5) (Fig. 3). Li et al. (2015) and Song et al. (2016)

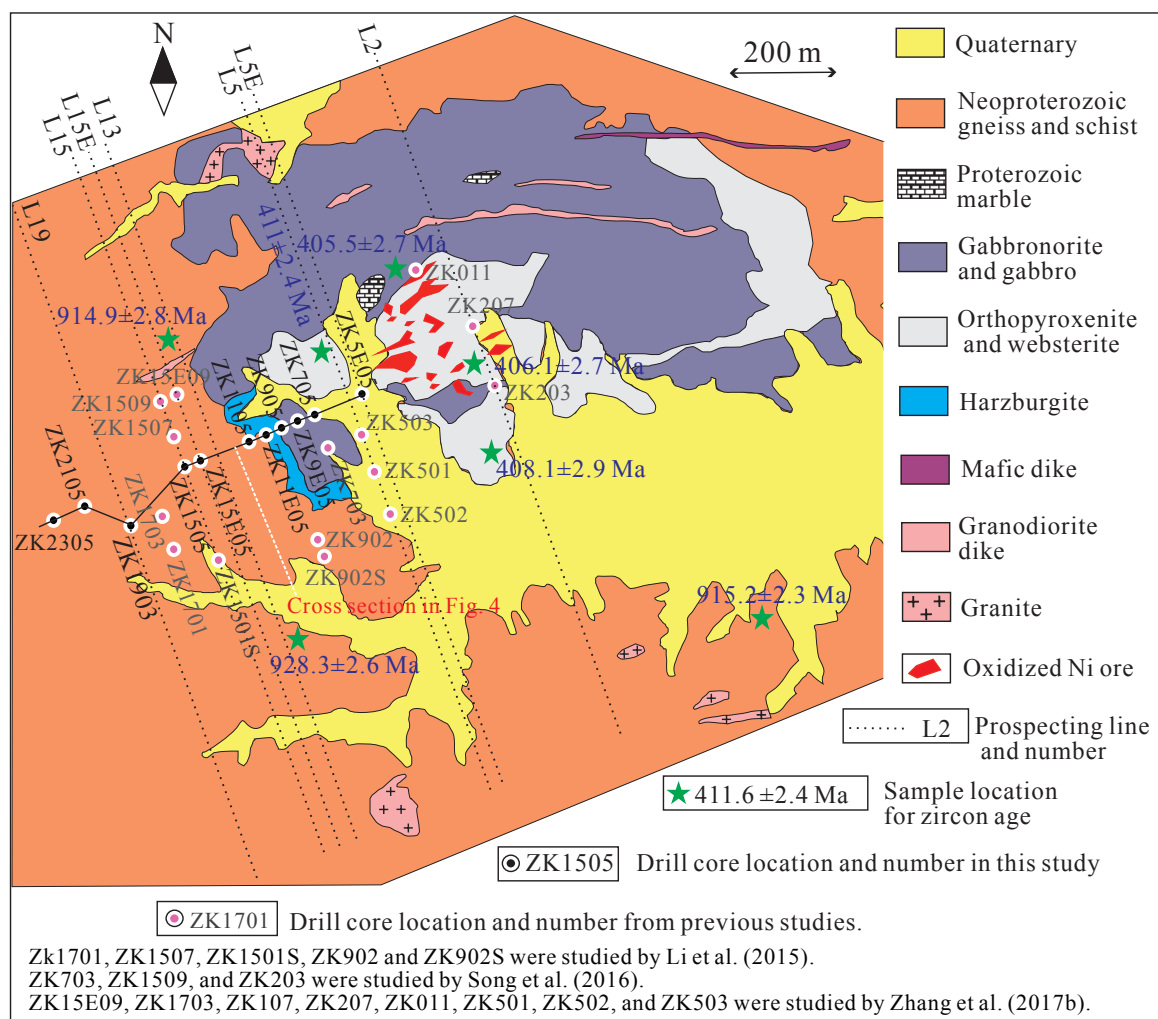


Fig. 3. Geological map of the Xiarihamu I mafic-ultramafic complex (modified from No. 5 Geological and Mineral Survey Institute of Qinghai Province, 2014). The data are from the following references: Neoproterozoic granitic gneiss ages (Gan, 2014; Wang et al., 2016), Xiarihamu I mafic-ultramafic complex age (Li et al., 2015; Song et al., 2016).

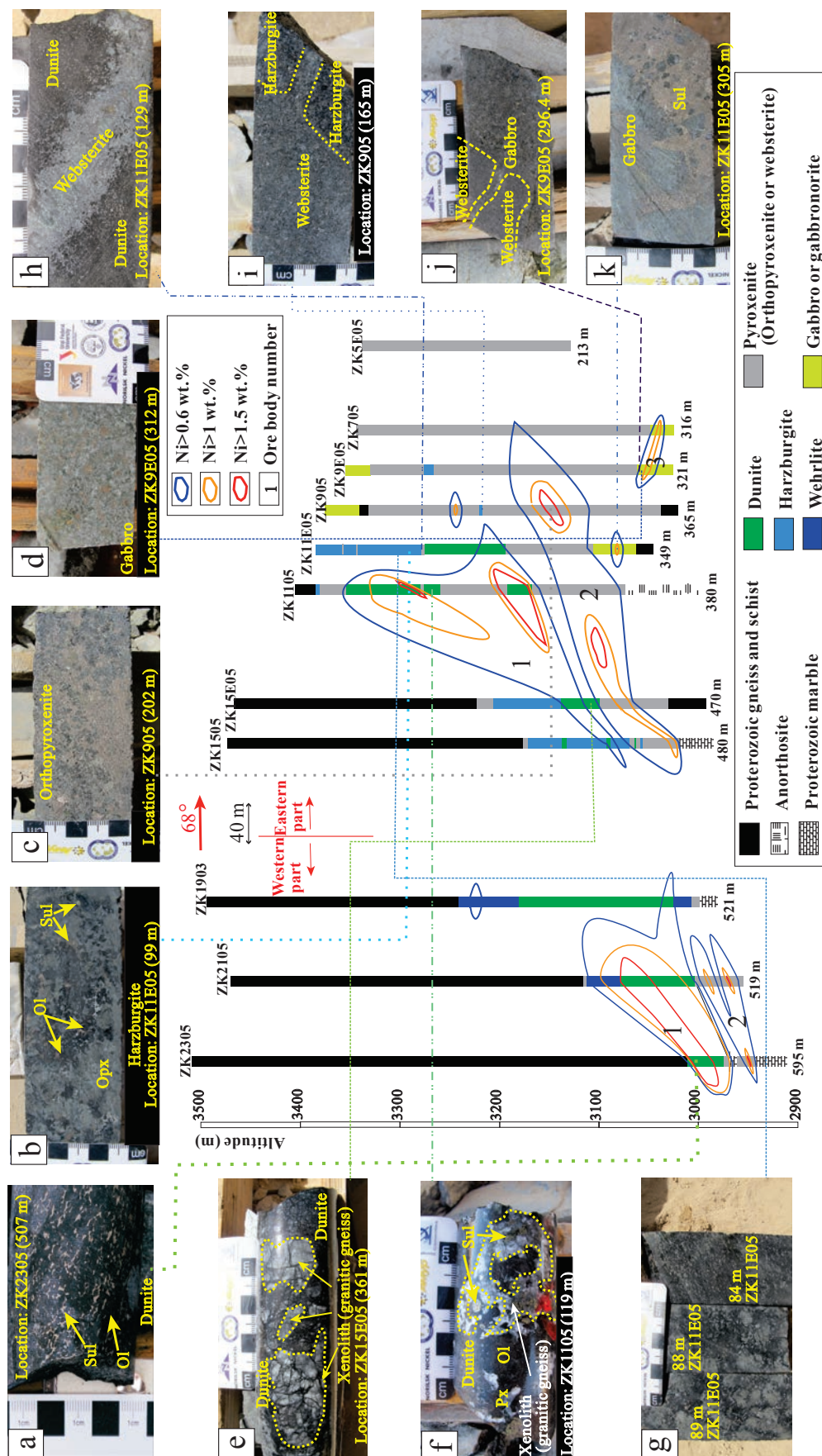


Fig. 4. Lithofacies distribution and orebodies from sampling locations in a cross section of the Xiarihamu Ni deposit. a. Dunite with strongly disseminated sulfide, 507 m, drill core ZK2305. b. Harzburgite with disseminated sulfide, 99 m, drill core ZK11E05. c. Websterite with massive sulfide, 202 m, drill core ZK905. d. Gabbro with disseminated sulfide, 312 m, drill core ZK9E05. e. Granitic gneiss xenoliths in the dunite, 361 m, drill core ZK15E05. f. Granitic gneiss xenoliths in the dunite, with some pyroxene and sulfides occurring between the olivine and the xenolith, 119 m, drill core ZK11E05. g. Harzburgite, the olivine grain size and orthopyroxene content gradually increase from top to bottom, drill core ZK11E05. h. Websterite cutting through dunite, 129 m, drill core ZK11E05. i. Websterite intruded into harzburgite, 165 m, drill core ZK905. j. Gabbro intruded into websterite, 296.4 m, drill core ZK9E05. k. Massive ore intruded into gabbro, 305 m, drill core ZK11E05.

have obtained ages of 411.6 ± 2.4 and 406.1 ± 2.7 Ma, respectively, for the websterite. The SHRIMP zircon U-Pb age of the gabbro between L2 and L5 is 405.5 ± 2.7 Ma (Fig. 3; Song et al., 2016), whereas the laser ablation-inductively coupled plasma-mass spectrometry (LA-ICP-MS) zircon U-Pb age of the gabbro from the drill core ZK1905 (Fig. 3) at the depth of 430 m is 431.3 ± 2.7 Ma (Li et al., 2015; Fig. 3). Hence, the gabbro most likely formed in two stages, and the ultramafic

body (i.e., the 406–411 Ma websterite) is 20 Ma younger than the older gabbro (431.3 ± 2.7 Ma).

We have logged 11 drill cores from one typical cross section (Figs. 3, 4) at a 1:100 scale. Based on detailed field observations and an analysis of 110 representative thin sections, the lithofacies distribution and crosscutting relationships have been determined, as shown in Figures 4 and 5. Figure 4 clearly shows that the mafic-ultramafic complex is narrow

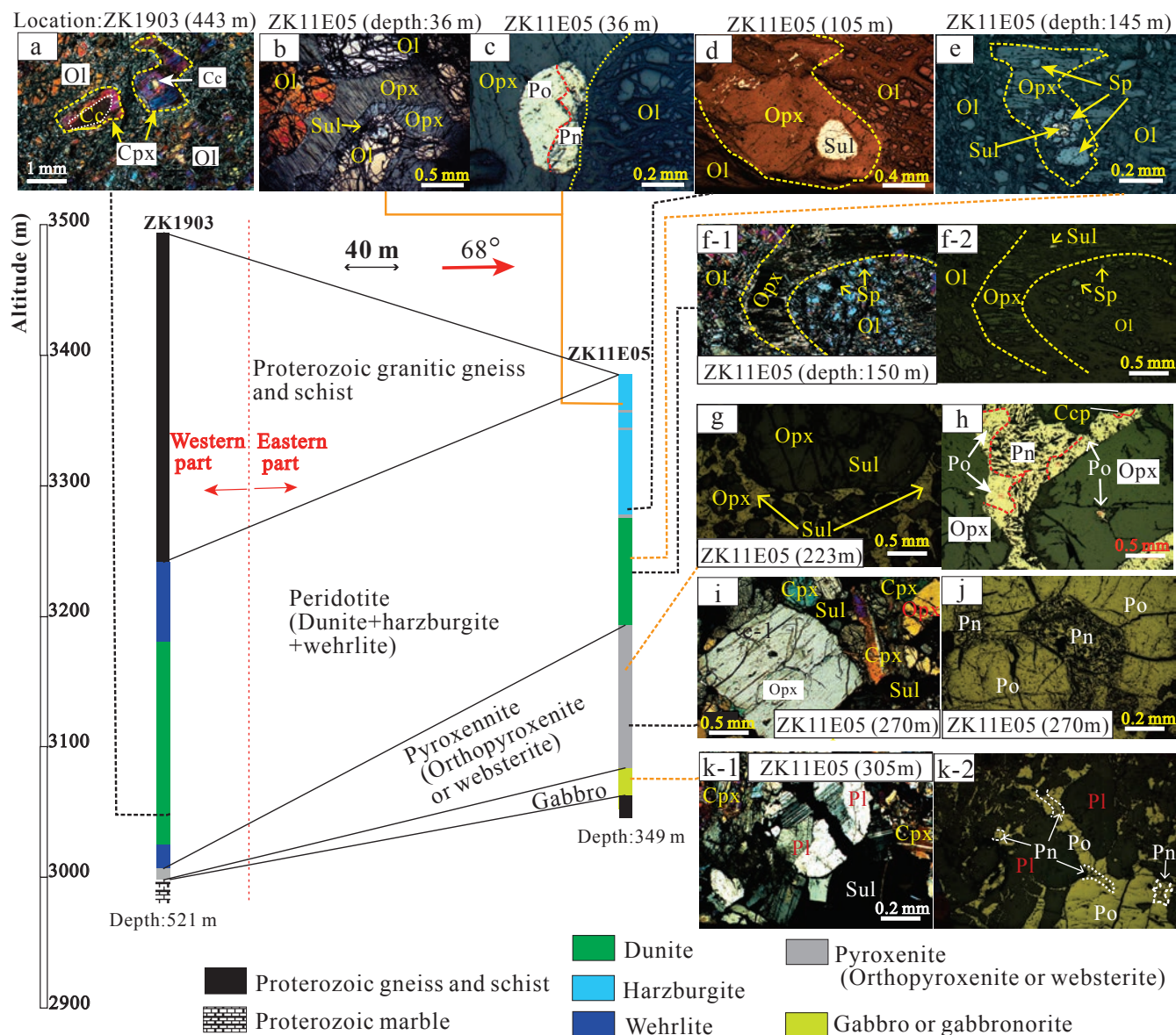


Fig. 5. Photomicrographs of sampling locations in main rock types. The fields of view in i-1 and i-2 are the same, but i-1 is under transmitted light and i-2 is under reflected light. Cc = calcite, Cpx = clinopyroxene, Ol = olivine, Opx = orthopyroxene, Pn = pentlandite, Po = pyrrhotite, Pl = plagioclase, Sp = spinel, and Sul = sulfide. a. Small calcite (marble) xenoliths in dunite, 443 m, drill core ZK1903. b. and c. Harzburgite, 36 m, drill core ZK11E05; the sulfide is present at the contact between olivine and orthopyroxene. d. Harzburgite in which the sulfides are located in the Opx, 105 m, drill core ZK11E05. e. Dunite: Sp is present outside the Ol; the Sp has sulfide inclusions, and one Sp grain is located in an Opx grain, 145 m, drill core ZK11E05. f. Dunite: Sp is present at the margins of Ol, and sulfide is located in the Opx grain, 150 m, drill core ZK11E05. g. and h. Websterite with coarse-grained Opx and fine-grained Opx; the sulfide is mainly located in the interstitial space between olivine and orthopyroxene, while minor sulfides are enclosed by orthopyroxene, 223 m, drill core ZK11E05. i. and j. Websterite with coarse-grained Opx and fine-grained Opx, the sulfide is mainly located in the interstitial space between orthopyroxene grains, 270 m, drill core ZK11E05. k. Massive ore intruded into the gabbro, and the massive sulfide cut through the plagioclase, 305 m, drill core ZK11E05.

in the western part and wider in the eastern part. The upper portion of the country rock is Proterozoic granitic gneiss, and the lower portion of the country rock is marble in the western part and Proterozoic gneiss in the eastern part. Small amounts of sulfide are locally present in the marble and granitic gneiss. For example, in drill core ZK2709, the marble contains less than 0.1 wt % S and is in the form of pyrite with a crystal size of 10 to 30 μm . In drill core ZK2105, the granite gneiss located at 216 m is less than 0.1 wt % S in the form of pyrite with a crystal size of 50 to 100 μm .

The rock types in the cross section include dunite, harzburgite, wehrlite, pyroxenite, and gabbro (Fig. 4). We divide this cross section into two parts: the western part and the eastern part. The western part includes drill cores ZK2305, ZK2105, and ZK1903. The eastern part includes drill cores ZK1505, ZK15E05, ZK1105, ZK11E05, ZK905, ZK9E05, ZK705, and ZK5E05 (Fig. 4).

In the western part of the cross section (Fig. 4), the thickness of the dunite decreases from 156 m in drill core ZK1903 to 36 m in ZK2305. From drill core ZK2305 to ZK1903, the rock overlying the dunite is wehrlite. The thickness of the wehrlite gradually decreases from 80 m in ZK1903 to 0.5 m in ZK2305 (Fig. 4). In addition, a wehrlite body 9 m thick is located below the dunite in drill core ZK1903. Below the dunite and wehrlite is pyroxenite with a thickness ranging from 14 to 49 m.

In the eastern part of the cross section (Fig. 4), the thickness of the dunite in drill core ZK1105 reaches a maximum of 87 m and gradually decreases to 5 m in drill core ZK1505 and 3 m in drill core ZK905. Harzburgite is located above dunite in drill cores ZK1505, ZK15E05, and ZK11E05 in direct contact with the Proterozoic granitic gneisses (Fig. 4). The dunite is underlain by pyroxenite, which increases in thickness from 50 m in drill core ZK 1505 to 265 m in drill core ZK705 (Fig. 4).

Based on the host rocks, the orebodies can be divided into three zones (Fig. 4): (1) ore zones located in dunite, referred to as the No. 1 orebody; (2) ore zones in pyroxenite (orthopyroxenite and websterite), referred to as the No. 2 orebody; and (3) ore zones in gabbro, referred to as the No. 3 orebody. The high-grade Ni ore (Ni > 1.5%) in the No. 1 orebody is mainly located in the lower portion of the dunite in drill cores ZK1505 to ZK1105 and in the upper portion of the dunite in drill cores ZK 2305 to ZK2105 (Fig. 4). The No. 2 orebody is situated in the central portion of the pyroxenite (orthopyroxenite or websterite) in drill cores ZK905 and ZK9E05 or the lower portion of the pyroxenite in drill cores ZK1505 and ZK11E05 (Fig. 4).

Petrology and crosscutting relationships

The dunites in different parts of the cross section differ significantly in terms of modal mineralogy, especially sulfide content. The dunite in drill core ZK2305 contains 10 to 15% disseminated sulfide (Fig. 4a), whereas the upper portion of the dunite in drill core ZK11E05 contains almost no sulfide or sparsely disseminated sulfide (Fig. 4h). The olivine content in dunite in drill cores ZK11E05 and ZK1903 is approximately 90%. The pyroxene content is nearly 7%, and the pyroxene grains are present as interstitial crystals (Fig. 5a, f-1, f-2). The pyroxene in the dunites varies in that the pyroxene in drill

core ZK11E05 is orthopyroxene (Fig. 5f-1, f-2), whereas the pyroxene in drill core ZK1903 is clinopyroxene (Fig. 5a). The spinel content in the dunites is 1 to 2% and occurs mainly along the edges of olivine grains (Fig. 5f-1, f-2) or as interstitial material between olivine grains (Fig. 5e). Therefore, olivine was the earliest phase to crystallize, followed by spinel. Some spinel contains sulfide inclusions (Fig. 5e). At depths of 361 m in drill core ZK15E05 and 119 m in ZK13E05, granitic gneiss xenoliths are observed in the dunite (Fig. 4e-f), and pyroxene and sulfide occur at the contact between the olivine and the granitic gneiss. Furthermore, at a depth of 443 m in drill core ZK1903, we observe small calcite (marble) xenoliths wrapped in the clinopyroxene in the dunite (Fig. 5a).

The typical harzburgite contains 2% disseminated sulfide (Fig. 4b), 45% olivine, 45% orthopyroxene, 2 to 3% clinopyroxene, and 1 to 2% spinel (Fig. 5b-d). Sulfides are mainly pentlandite and pyrrhotite (Fig. 5c) and are well distributed and have a disseminated texture. The sulfide is present at the contact between olivine and orthopyroxene (Fig. 5c) or in the orthopyroxene (Fig. 5d). With increasing depth in the harzburgite in drill core ZK11E05, the grain sizes of the olivine and orthopyroxene increase, and the proportion of orthopyroxene gradually increases (Fig. 4g).

The websterite contains sparse disseminated sulfide or massive sulfide (Figs. 4c, 5g-j). The orthopyroxene in websterite has two distinct crystal sizes: larger grains approximately 1 to 1.5 mm in diameter and smaller grains 0.2 to 0.5 mm in diameter (Fig. 5g, i). The sulfides are located mainly in the interstitial space between orthopyroxene grains (Fig. 5g, i), but minor sulfides are enclosed by orthopyroxene (Fig. 5i). The sulfides (0.5–5%) in the websterite are mainly pyrrhotite and pentlandite (Fig. 5h, j), with minor chalcopyrite (Fig. 5h). The upper portion of the websterite has more chalcopyrite than the lower portion in drill core ZK11E05 (Fig. 5h).

Gabbro is present at the bottom of drill core ZK11E05, and a portion of this gabbro contains 5 to 6% disseminated sulfides (Fig. 4d) and massive sulfides (Fig. 4k). There are fragments of angular gabbros where the gabbro is in contact with the massive orebodies (Fig. 4k), and the massive sulfide has cut through the plagioclase (Fig. 5k-1, k-2). The massive ore consists of main pyrrhotite with minor pentlandite (Fig. 5k-2).

No crosscutting relationships exist among the dunite, harzburgite, and wehrlite, suggesting that these units formed sequentially during the same stage of magmatic evolution. In contrast, a websterite vein cuts through the dunite and harzburgite (Fig. 4h-i), indicating that the websterite formed later than the dunite and harzburgite. Similarly, a gabbro vein intruded into the websterite (Fig. 4j), suggesting that the gabbro formed slightly later than the websterite. Based on zircon dating results, one of the gabbros formed 20 m.y. before the websterite (Li et al., 2015). Therefore, this paper concludes that the intrusion sequence is as follows: gabbro at 431 Ma; dunite-harzburgite-wehrlite, websterite or orthopyroxenite at 405 to 411 Ma; and gabbro at the bottom of drill core ZK11E05. In the cross section of this study, we focused on the latter three stages: (Figs. 4, 5): (1) the peridotite, including dunite, harzburgite, and wehrlite, located at the upper section; (2) the pyroxenite, including websterite and orthopyroxenite, located at the middle section; and (3) the gabbro, located at the bottom section (Fig. 5). In this study, the samples were

mainly collected from two typical drill cores: ZK11E05 and ZK1903.

Analytical Methodology

Major element geochemistry of minerals

The equilibration of the olivine with trapped sulfide liquids could change the olivine composition significantly (Duke and Naldrett, 1978; Fleet and Macrae, 1988). Therefore, we selected the samples from drill cores ZK11E05 and ZK1903, as most of the peridotite in these two drill cores have sparsely disseminated sulfide or no sulfide. This minimizes the effect of sulfides on minerals such as olivine. Mineral compositions were determined using an electron microprobe at Chang'an University. The analytical conditions included an energy of 15 kV, a beam current of 20 nA, a beam diameter of 1 to 5 μm and a peak-counting time of 20 s. The standard materials used in the analysis were as follows: albite, K-feldspar, hematite, forsterite, xenotime, pyrophanite, corundum, wollastonite, Cr spinel, kyanite, pyrophanite, metallic nickel, and metal zinc, respectively. All materials were selected according to the National Standard of the People's Republic of China (1998). The lower limit of detection for all of the elements was 0.01%. The analytical relative error was $\pm 2\%$. The results of the mineral compositions and sample location are listed in Supplementary Table 1.

Ni, Cu, S, and PGE geochemistry

Whole-rock Ni contents were compiled from the No. 5 Geological and Mineral Survey Institute of Qinghai Province. These data included 166 samples from drill core ZK11E05 and 131 samples from drill core ZK1903. The location and the results have been listed in Supplementary Table 2.

In addition, 12 and five samples were collected from drill cores ZK11E05 and ZK1903, respectively, for Ni, Cu, S, and PGE analyses. The location and the results have been listed in Table 1. Those samples were crushed in steel jaws to ~ 10 mesh powder, and ~ 200 g of this fraction was then ground to ~ 200 mesh powder using a tungsten carbide ring mill. Whole-rock Cu and Ni contents of samples were measured via inductively coupled plasma atomic emission spectroscopy (ICP-AES), with lower limits of detection of ~ 0.01 wt %, at ALS Chemex (Guangzhou) Co. Ltd. Whole-rock S contents were measured using a gravimetric method and IR absorption (IR08) by the infrared sensor at 1,350°C with a lower limit of detection of ~ 0.01 wt %. The analytical precision was $\pm 8\%$ of the amount present for S and $\pm 3\%$ for Ni and Cu. The PGE concentrations were determined at Nanjing Focu MS Technology Co. Ltd. by isotope dilution (ID)-ICP-MS, using an improved digestion technique with beakers sealed in a stainless steel pressure bomb (Qi et al., 2011). The total procedural blanks varied from 0.009 ng (Ir) to 0.033 ng (Pd), and the lower limits of detection ranged from 0.004 ppb (Ir) to 0.012 ppb (Pd) (Qi et al., 2011). The precision and accuracy were monitored via analyses of the standards WGB-1 and UMT-1 as well as the data of four duplicate samples listed in Table 1.

Whole-rock Sr isotope compositions

High-precision Sr isotope measurements were carried out at Nanjing Focu MS Technology Co. Ltd. Fifty milligrams

Table 1. PGE (Ir, Ru, Rh, Pt, and Pd), Cu, Ni, and S Contents of the Rocks from Drill Cores ZK11E05 and ZK1903

Drill core	Depth (m)	Sample description	S (%)	Ni (total) (ppm)	Ni (in sulfide) (ppm)	Ir (ppb)	Ru (ppb)	Rh (ppb)	Pt (ppb)	Pd (ppb)	Cu (ppm)	Σ PGE (ppb)	Pd/Ir	Cu/Pd	Ni _{100%} (ppm)	Pd _{100%} (ppb)	PGE _{100%} (ppb)
ZK11E05	89	Harzburgite	0.65	2600	1330	0.19	0.12	0.62	0.66	1.68	440	3.27	8.7	262,133	7.73	94.3	184
ZK11E05	99	Harzburgite	2.87	7900	6987	0.57	0.66	0.69	0.37	6.10	4150	8.39	10.7	680,106	9.05	78.6	108
ZK11E05	114	Dunite	0.89	5800	3837	0.82	0.99	0.88	7.52	1.63	550	11.84	2.0	338,135	15.67	64.0	466
ZK11E05	140	Dunite	0.65	3900	2125	0.58	0.94	1.05	6.81	1.98	540	11.37	3.4	272,718	12.07	107.3	616
ZK11E05	171	Dunite	1.31	4900	2386	0.67	0.81	1.21	7.95	2.85	560	13.48	4.3	197,000	6.93	80.0	379
ZK11E05	192	Websterite	1.55	3300	2967	0.15	0.14	0.66	6.53	2.41	790	9.89	16.3	327,791	7.27	59.0	241
ZK11E05	213	Websterite	1.77	3100	2938	0.15	0.15	0.84	0.33	2.34	620	3.81	15.7	264,691	6.34	50.5	82
ZK11E05	228	Websterite	3.37	7500	5200	0.34	0.51	1.53	0.53	4.19	1120	7.10	12.3	267,421	8.42	47.0	80
ZK11E05	238	Websterite	2.16	5200	5000	0.23	0.33	2.98	0.59	3.09	680	7.22	13.5	219,999	9.08	54.0	126
ZK11E05	259	Websterite	2.89	5700	5511	0.28	0.39	1.01	1.35	3.92	1700	6.95	13.9	433,543	7.23	51.4	91
ZK11E05	263	Websterite	2.55	5800	5800	0.31	0.35	1.25	0.24	6.97	980	9.12	22.8	140,632	8.59	103.2	135
ZK11E05	270	Websterite	4.78	9300	9300	0.55	0.91	1.30	0.61	5.32	660	8.69	9.7	124,132	7.42	42.4	69
ZK11E05	310	Gabbro	3.54	5200	5200	0.21	0.19	1.48	1.07	2.35	6300	4.31	11.0	2,676,285	5.53	25.0	46
ZK11E05	320	Gabbro	1.15	2500	2500	0.26	0.37	1.00	2.27	1.13	710	5.03	4.3	629,963	8.20	37.0	165
ZK1903	266	Webhlite	1.07	6800	5460	0.05	0.03	0.40	2.64	4.10	1570	7.23	78.3	382,568	18.11	133.3	235
ZK1903	461	Dunite	3.25	11900	9547	0.23	0.25	1.18	0.05	3.05	970	4.75	13.4	318,074	10.98	34.6	54
ZK1903	465	Dunite	0.99	7000	4889	0.10	0.22	1.01	1.32	1.80	840	4.45	17.5	466,158	17.71	63.0	156
Average			2.08														
Recommended value		UMT-1				8.80	10.90	9.50	128.00	106.00							
Recommended value		WPR-1				13.50	22.00	13.40	285.00	235.00							
Measured value		UMT-1				7.81	9.54	10.24	121.94	104.56							
Measured value		WPR-1				17.77	22.46	12.92	296.63	233.61							

of rock powder was decomposed in high-pressure polytetrafluoroethylene (PTFE) bombs. Strontium was purified from the same digestion solution. An exchange column containing Bio-Rad AG50W×8 and Sr Spec resin was used to separate Sr, REE, and Pb from the sample matrix. The Sr-bearing elution was dried down and redissolved in 1.0 ml 2wt % HNO₃. Small aliquots of each were analyzed using an Agilent Technologies 7700x quadrupole multiple collector-inductively coupled plasma-mass spectrometer (MC-ICP-MS) (Hachioji, Tokyo, Japan) to determine the exact contents of Sr available. Diluted solution (50 ppb Sr doping with 10 ppb Tl) was introduced into an Nu Instruments *Nu Plasma II* MC-ICP-MS (Wrexham, Wales, UK) by a Teledyne Cetac Technologies *Aridus II* desolvating nebulizer system (Omaha, Nebraska, United States). Raw data of isotopic ratios were corrected for mass fractionation by normalizing to $^{86}\text{Sr}/^{88}\text{Sr} = 0.1194$ for Sr with the exponential law. International isotopic standards (NIST SRM 987 for Sr) were periodically analyzed to correct instrumental drift. Geochemical reference materials of USGS BCR-2, BHVO-2, AVG-2, and RGM-2 were used as a quality control. After the above chemical pretreatment and mass spectrometry, the results of the Sr isotope ratios were consistent with the reported values in the error range (Wei et al., 2013).

In situ S isotope analysis of sulfides

The ^{34}S and ^{32}S values were measured in situ on a ThermoFisher NEPTUNE Plus MC-ICP-MS in combination with a Geolas 2005 excimer Ar-F laser ablation system (Lambda Physik, Gottingen, Germany) by the method of Fu et al. (2016) at the State Key Laboratory of Geological Processes and Mineral Resources, China University of Geosciences in Wuhan, China. Before the analysis, the sulfide minerals were identified in the probe tablets and the sulfides with particle size >70 μm and a smooth surface were selected for analysis. Pyrite PPP-1 was used as a standard. The S isotope values of pyrite were determined with spot sizes of 32 μm using the single-spot ablation mode, whereas the S isotope values of pyrrhotite and chalcopyrite were analyzed with 32- μm spot sizes using the line scan ablation mode. The external precisions were 0.15 to 0.45‰ (2s) for $\delta^{34}\text{S}_{\text{V-CDT}}$ with an accuracy of 0.15‰ for $\delta^{34}\text{S}_{\text{V-CDT}}$ (Fu et al., 2016).

Trace element analyses of zircon in the Xiarihamu eclogite

Five eclogite samples were collected near the Xiarihamu Ni deposit (Fig. 3). All of the samples were collected within an area of 10 m². Trace element analyses of zircon from the eclogite were conducted synchronously by LA-ICP-MS at the Wuhan Sample Solution Analytical Technology Co. Ltd. Laser sampling was performed using a GeolasPro laser ablation system that consists of a COMPexPro 102 ArF excimer laser (wavelength of 193 nm and maximum energy of 200 mJ) and a MicroLas optical system. An 80-mJ laser energy was used in this study. An Agilent 7700e ICP-MS instrument was used to acquire ion-signal intensities. An Agilent standard tuning solution was used to autotune the best conditions of the instrument, including intensity, oxide production, and stability. Helium was used as a carrier gas and argon as a make-up gas, and these gases were mixed via a T connector before entering the ICP. A wire signal smoothing device is

included in this laser ablation system, by which smooth signals are produced even at very low laser repetition rates down to 1 Hz (Hu et al., 2012). The spot size and frequency of the laser were set to 32 μm and 5 Hz, respectively. Each analysis incorporated a background acquisition of approximately 20 to 30 s (gas blank) followed by a 50-s data acquisition from the sample. NIST610 was used as external standards for the trace element analysis. The trace element composition analysis of zircon was calibrated with Si as the internal standard. An Excel-based software ICPMSDataCal10 was used to perform off-line selection and integration of background and analyzed signals, time-drift correction, and quantitative calibration for trace element analysis and U-Pb dating (Liu et al., 2008, 2010). Concordia diagrams and weighted mean calculations were made using Isoplot/Ex_ver3 (Ludwig, 2003).

Results

Major element compositions of minerals

All of the major element results for the minerals are listed in Supplementary Table 1.

Mineral compositions at drill core ZK11E05: The Fo values of olivine in the harzburgite increase slightly from top to bottom, and except for a value of 85.3 at 79 m, the Fo values range from 86.8 to 88.4 (Fig. 6a). The NiO content of the olivine in the harzburgite increases from 0.16% at the top (30 m) to 0.32% at the bottom (93 m) (Fig. 6a). The Fo values of the dunite olivine in drill core ZK11E05 increase from 86.4 at the bottom (190 m) to 90.8 at the top (135 m) (Fig. 6a). The olivine NiO content gradually increases from 0.10% at 190 m to 0.41% at 135 m (Fig. 6a). The average SiO₂ and CaO contents of the dunite olivine in drill core ZK11E05 are 40.52 and 0.03%, respectively (Supplementary Table 1).

An olivine crystal whose composition trends from Mg to Fe rich toward the rim is considered to exhibit normal zoning, whereas the converse pattern is considered reverse zoning. The olivine at 45 and 145 m from the upper marginal zone and central zone of the peridotite (including the harzburgite and dunite), respectively, show normal zoning in the core but reverse zoning in the rim (Fig. 7a). In the upper marginal zone of the peridotite, the highest Fo value in the rim is lower than that in the core. In contrast, in the central zone, the highest Fo value in the rim is nearly the same as that in the core, and the reverse curve at the central zone is steeper than that in the upper zone. The olivine at 179 m from the lower zone exhibits normal zoning (Fig. 7a).

The Cr number [atomic Cr/(Cr + Al)] and FeO values of the spinel from drill core ZK11E05 tend to decrease downward in the peridotite facies (including the dunite and harzburgite; Fig. 6b). The Cr[#] and FeO shift from 0.61 and 37.05%, respectively, at the top to 0.26 and 18.75%, respectively, at the bottom. (Fig. 6b). The Fe³⁺ content of spinel was calculated using the method of Droop (1987). The calculation process has been provided in Supplementary Table 3. The Fe³⁺/ΣFe ratios (i.e., atomic, Fe³⁺/[Fe²⁺ + Fe³⁺]) of the spinel in drill core ZK11E05 range from 0.05 to 0.30.

The Mg number [atomic Mg/(Mg + Fe²⁺)] value of the orthopyroxene in the peridotite facies of drill core ZK 11E05 ranges from 0.88 to 0.89. No obvious trend with depth was found. The orthopyroxene in the websterite can be divided

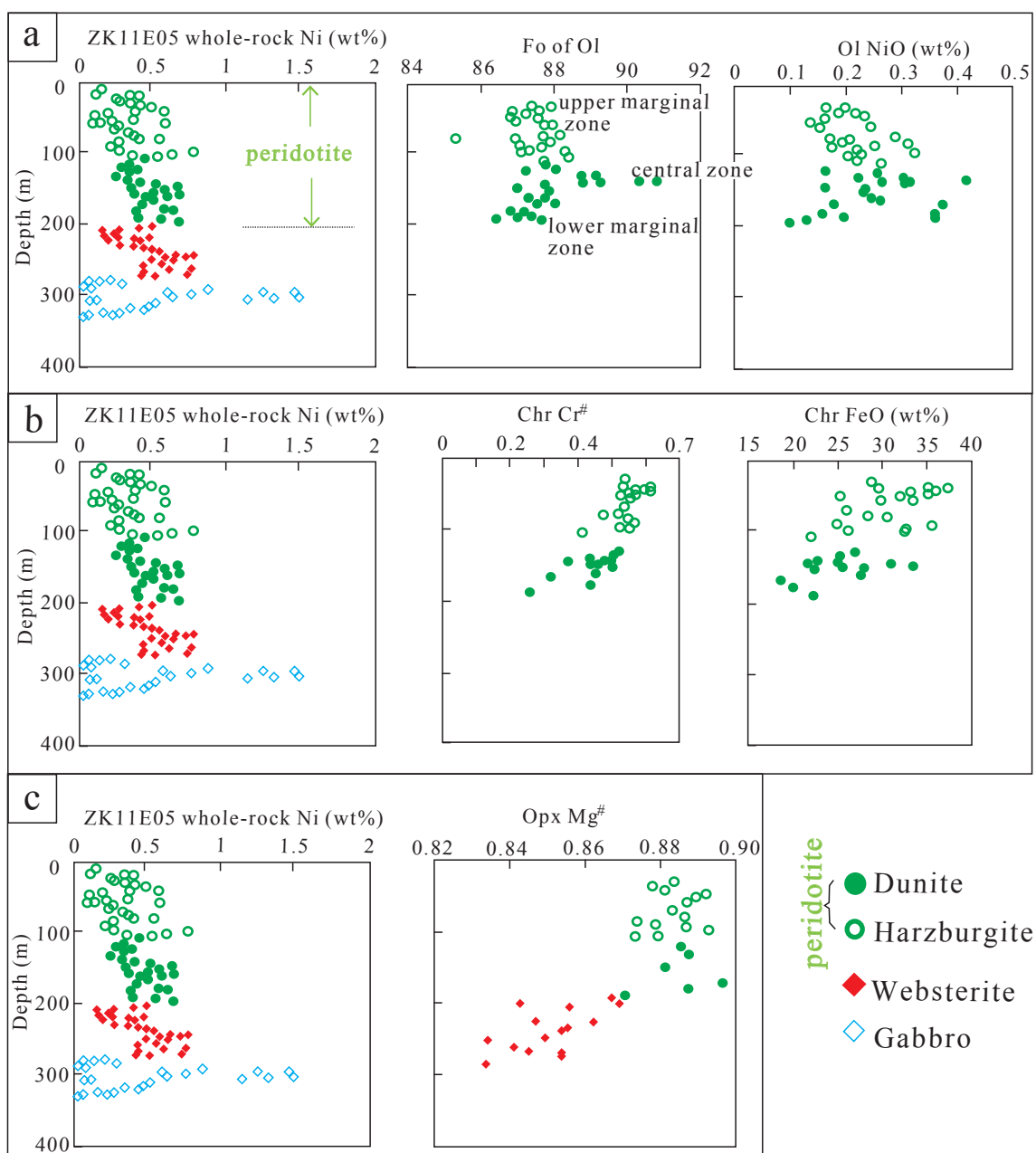


Fig. 6. The relationship between the Ni content and the composition of olivine (a), spinel (b), and orthopyroxene (c) in drill core ZK11E05.

into two categories according to the crystal size (Fig. 5g, i). The composition of the fine orthopyroxene is probably easily affected by the coexisting sulfide; therefore, we report only the composition of the coarse orthopyroxene. The Mg number value of the orthopyroxene in the websterite decreases rapidly from 0.87 at the top to 0.83 at the bottom (Fig. 6c). The orthopyroxene in the websterite exhibits normal zoning (Fig. 7b).

Mineral compositions at drill core ZK1903: The olivine Fo and NiO values decrease from the central part upward and downward (Fig. 8a). The Fo value decreases from 89.2 at 351 m (central part) to 83.8 at 266 m (the upper marginal zone) to 85.6 at 476 m (the lower marginal zone) (Fig. 8a).

An Fo value of 83.0 exists at the lower marginal zone, but this data plot far from the trend line. The NiO content in the olivine from the central zone is 0.32% at 311 m, which gradually decreases to 0.18% at 473 m (Fig. 8a). The average SiO₂ and CaO contents of the dunite olivine in drill core ZK1903 are 40.05 and 0.04%, respectively (Supplementary Table 1).

The olivine at 258 and 346 m from the upper marginal zone and central zone of the peridotite (including the wehrlite and dunite) in drill core ZK1903, respectively, show normal zoning in the core but reverse zoning in the rim (Fig. 7c). In the upper marginal zone, one side of the rim shows normal zoning (Fig. 7c). The highest Fo value in the rim in the central zone of the peridotite is higher than that in the core. The olivine at

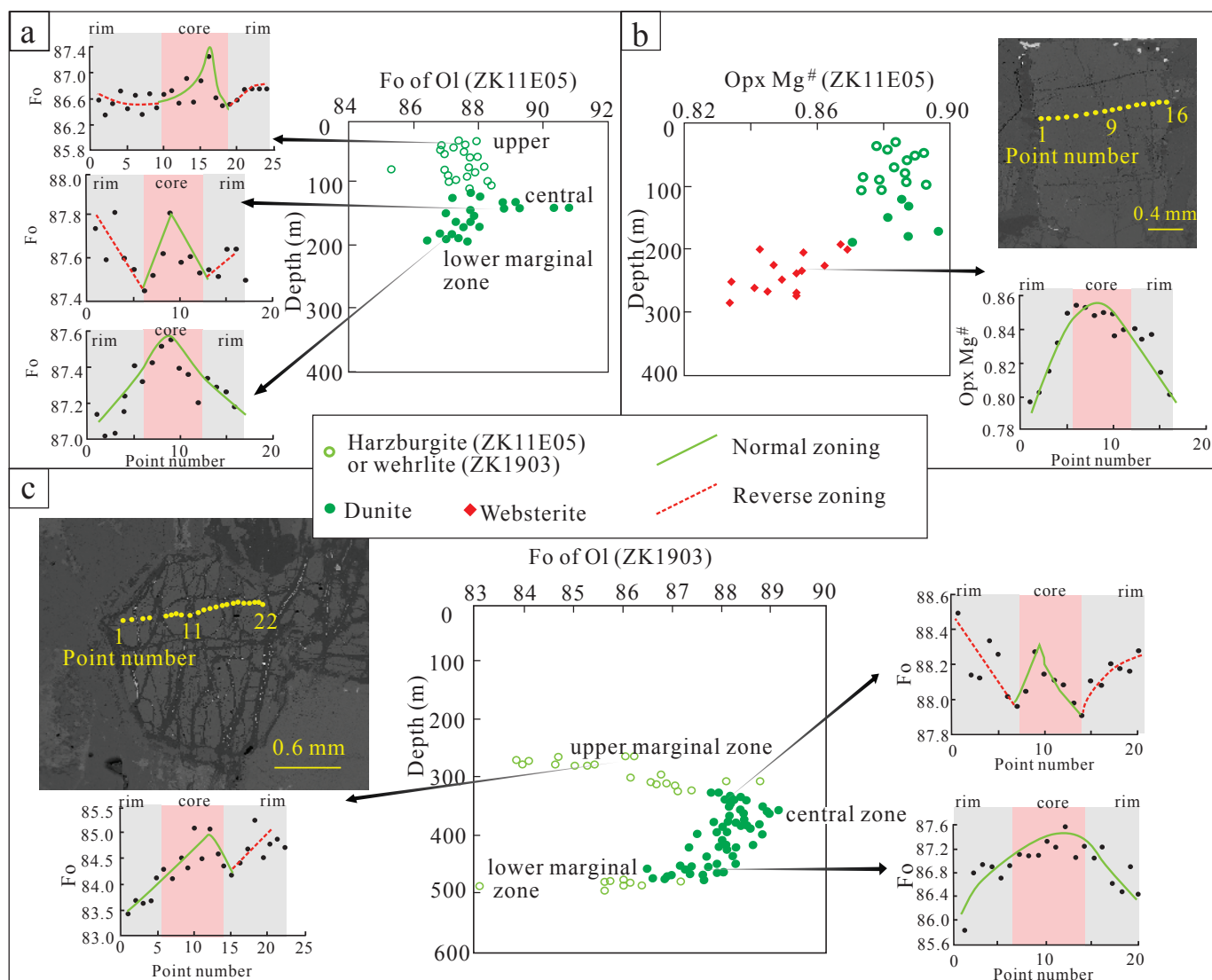


Fig. 7. The normal zoning and reverse zoning of olivine and orthopyroxene.

469 m from the lower marginal zone exhibits normal zoning (Fig. 7c).

The Cr number values of the spinel in drill core ZK1903 range from 0.15 to 0.53. Most of the values range from 0.3 to 0.5. Although the Cr number values do not show an obvious trend with depth, the FeO values decrease downward (Fig. 8b). This trend was also found in drill core ZK11E05 (Fig. 6b). The $\text{Fe}^{3+}/\Sigma\text{Fe}$ ratios of the spinel in drill core ZK1903 range from 0 to 0.20 (Supplementary Table 1).

The Mg number values of the clinopyroxene in drill core ZK1903 fluctuate, ranging from 0.84 to 0.94, but the trend exists where the value decreases from the central part when going upward or downward (Fig. 8c).

Ni, Cu, PGEs, and S contents

The whole-rock Ni content of the peridotite (including the dunite and harzburgite) in drill core ZK11E05 gradually increases from 0.11 wt % at 14.3 m at the top to 0.63 wt % at 183 m at the bottom (Fig. 6a). In the websterite, the Ni

content increases downward. The lowest Ni content at the top is 0.2 wt %, whereas the highest Ni content at the bottom of the websterite is 0.78 wt % at 273 m. The Ni content of the gabbro rapidly decreases from 1.45 wt % at 305 m to 0.019 wt % at 340 m at the bottom and to 0.2 wt % at 285 m at the top of the gabbro (Fig. 6a).

In Figure 8a, the whole-rock Ni contents at the bottom (453–471 m) of the dunite and in the upper marginal zone (253–267 m) of drill core ZK1903 suddenly increase to values greater than 0.5 wt %, and the samples at these two locations have disseminated sulfide or tense disseminated sulfide; however, from 268 to 452 m, the whole-rock Ni contents remain at nearly 0.2 wt %, and either no sulfide is present or its presence is rare.

In addition to the whole-rock nickel data compiled from our cooperative institute, we selected 14 samples from drill core ZK11E05 and three samples from drill core ZK1903 containing S > 0.65 wt % for PGE, Cu, and Ni analyses. The results and the location of the samples are listed in Table 1. In this

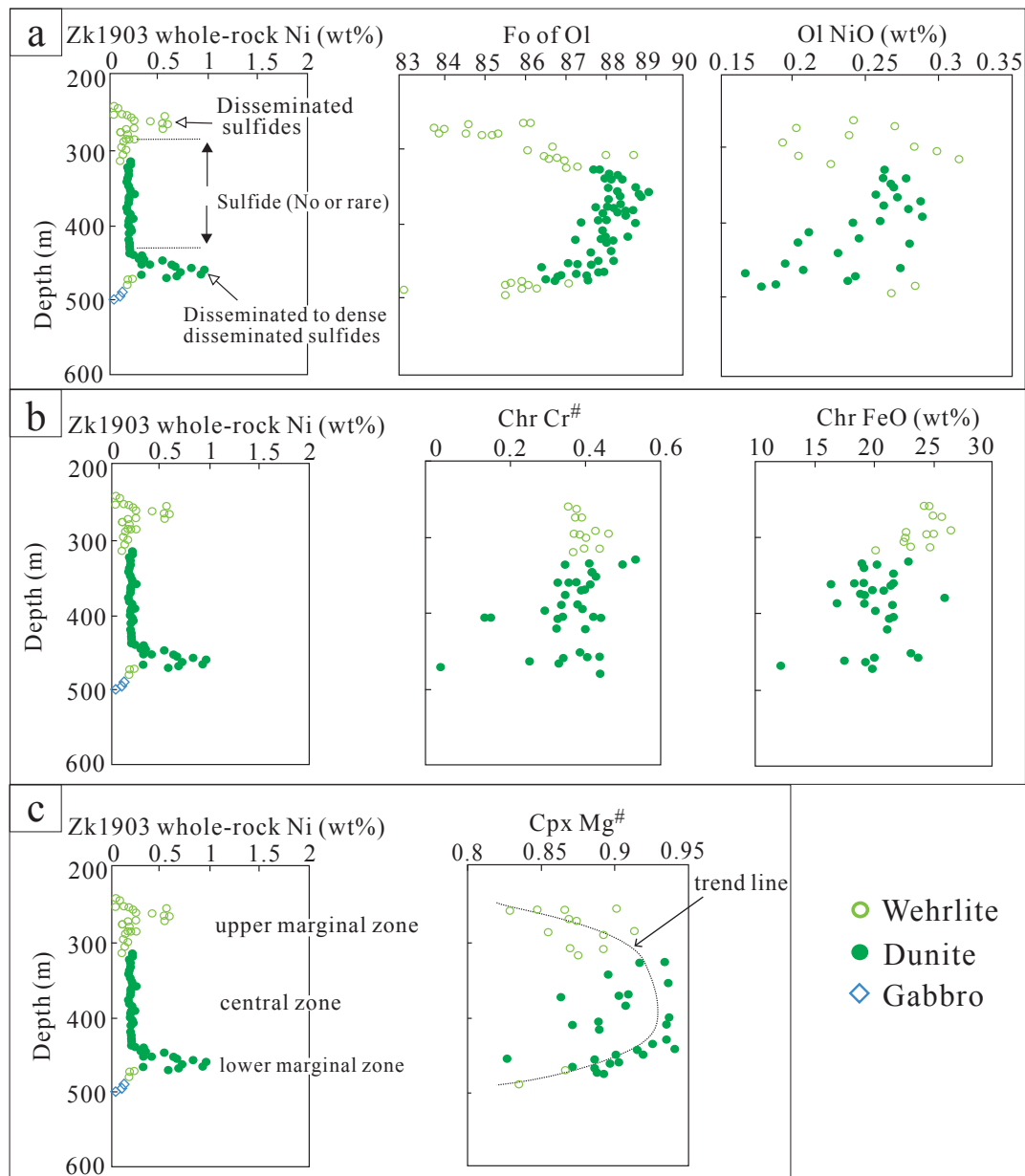


Fig. 8. The relationship between the Ni content and the composition of olivine (a), spinel (b) and clinopyroxene (c) in drill core ZK1903.

paper, the content of each PGE in 100% sulfide is calculated according to equation (1) (Barnes and Ripley, 2016):

$$C_{100\%Sul} = \frac{C_{wr} \times 100}{2.527 \times S + 0.3408 \times Cu + 0.4715 \times (Ni - Ni_{sil})}, \quad (1)$$

where $C_{100\%Sul}$ is the content of each PGE in the 100% sulfide; C_{wr} is the content (%) of the PGE element in the whole rock; S, Cu, and Ni are concentrations of these elements in the whole rock; and Ni_{sil} is the nickel content in silicates and oxides. In this paper, Ni_{sil} was calculated using equation (2):

$$Ni_{sil} = Ni_{Ol} \times \text{the olivine mass fraction} + Ni_{Sp} \times \text{the spinel mass fraction} + Ni_{Px} \times \text{the pyroxene mass fraction}. \quad (2)$$

The Ni contents in Ol-Sp-Px were averaged from the electron probe microanalysis (EPMA) results, and the mass fractions were estimated from the mineral volume fractions in thin sections.

In Table 1, $PGE_{100\%}$ is the sum of the contents of the five PGEs (Ir, Ru, Rh, Pt, and Pd) in the 100% sulfide. The whole-rock PGE contents and S contents of the peridotite (including dunite and harzburgite) in drill core ZK11E05 tend to increase downward (Fig. 9), except for one harzburgite sample at 99 m with 2.87 wt % S. $PGE_{100\%}$ and $Ni_{100\%}$ values reach their highest values near the location where the highest olivine Fo value occurs. The Pd/Ir and Cu/Pd ratios of the peridotite are 2.0 to 10.7 and 19,700 to 680,106, respectively, and no obvious trend was found upward or downward (Fig. 9, Table 1).

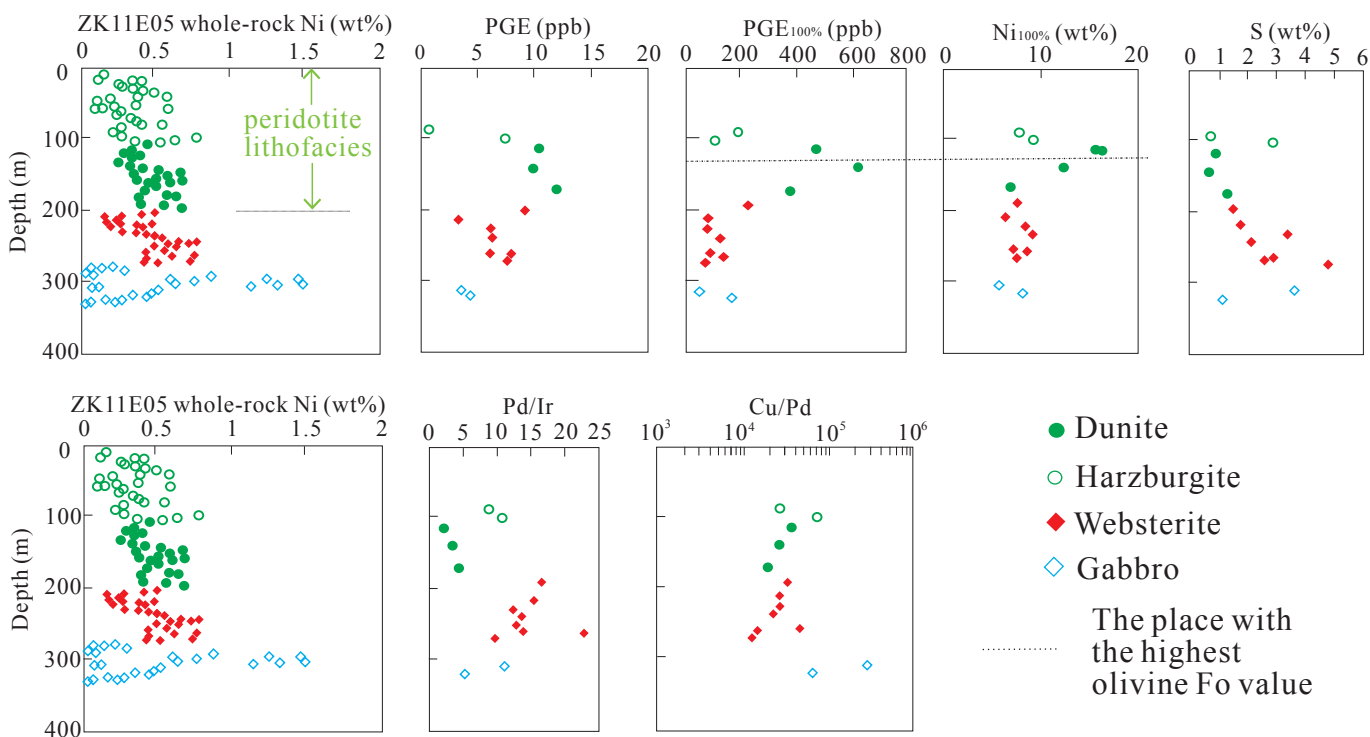


Fig. 9. The relationship between the Ni content and the PGE features and S contents.

The PGE, PGE_{100%}, and Ni_{100%} values of the websterite in drill core ZK11E05 are 3.81 to 9.85 ppb, 69 to 241 ppb, and 6.34 to 9.08 wt %, respectively. The S content of the websterite in drill core ZK11E05 increases downward, from 1.55 wt % at the top to 4.78 wt % at the bottom (Fig. 9). The Pd/Ir and Cu/Pd values tend to decrease downward (Fig. 9, Table 1). Pd/Ir ratios decrease gradually from 16.3 at 192 m to 9.7 at 270 m, except for one value of 22.8 at 263 m, while the Cu/Pd value decreases from 327,791 at the top 192 m to the 124,132 at the bottom 270 m, except for one 433,543 value at 259 m (Fig. 9, Table 1).

The PGE, PGE_{100%}, and Ni_{100%} values of the gabbro in drill core ZK11E05 are 4.31 to 5.01 ppb, 46 to 165 ppb, and 5.52 to 8.20 wt %, respectively (Fig. 9, Table 1). The S, Pd/Ir, and Cu/Pd values of these gabbros are 1.15 to 3.54, 4.4 to 11.0, and 629,963 to 2,676,285 wt %, respectively (Fig. 9, Table 1).

The PGE, PGE_{100%}, and Ni_{100%} values of three sample peridotite rocks in drill core ZK1903 are 4.45 to 7.23 ppb, 54 to 235 ppb, and 10.98 to 18.11 wt %, respectively (Table 1). The S, Pd/Ir, and Cu/Pd values of the above three sample peridotite rocks are 0.99 to 3.25, 13.4 to 78.3, and 318,074 to 466,158 wt %, respectively (Table 1).

Whole-rock Sr isotopes

The Sr isotope values of the rocks from drill cores ZK11E05 and ZK1903 are listed in Table 2 and Figure 10. The initial ⁸⁷Sr/⁸⁶Sr (⁸⁷Sr/⁸⁶Sr)_i values were calculated using a zircon U-Pb age of 411 Ma (Li et al., 2015).

The (⁸⁷Sr/⁸⁶Sr)_i value of the dunite in drill core ZK11E05 is 0.71064, which is slightly higher than the harzburgite (⁸⁷Sr/⁸⁶Sr)_i value of 0.70954 (Fig. 10a). The (⁸⁷Sr/⁸⁶Sr)_i values of the two websterite samples are 0.71037 at 259 m and 0.71020 at 283 m (Fig. 10a).

Table 2. Sr Isotope Values of the Rocks from Drill Cores ZK11E05 and ZK1903

Drill core	Depth (m)	Rock type	⁸⁷ Sr/ ⁸⁶ Sr	±2σ	Rb (ppm)	Sr (ppm)	Age (Ma)	⁸⁷ Rb/ ⁸⁶ Sr	(⁸⁷ Sr/ ⁸⁶ Sr) _i
ZK11E05	18	Harzburgite	0.715013	0.000004	10.3	31.9	411.0	0.93429	0.70954
ZK11E05	170	Dunite	0.714807	0.000003	3.2	13.0	411.0	0.71227	0.71064
ZK11E05	259	Websterite	0.712439	0.000003	5.1	41.8	411.0	0.35305	0.71037
ZK11E05	283	Websterite	0.713046	0.000003	6.7	39.8	411.0	0.48711	0.71020
ZK1903	258	Wehrlite	0.708932	0.000003	9.6	114.5	411.0	0.24261	0.70751
ZK1903	294	Wehrlite	0.710053	0.000003	7.3	90.3	411.0	0.23392	0.70868
ZK1903	361	Dunite	0.709044	0.000003	5.5	28.2	411.0	0.56435	0.70574
ZK1903	418	Dunite	0.714966	0.000003	4.2	35.3	411.0	0.34428	0.71295
ZK1903	461	Dunite	0.711593	0.000004	4.7	75.4	411.0	0.18037	0.71054
ZK1903	481	Wehrlite	0.711295	0.000003	9.2	78.0	411.0	0.34130	0.70930

Note: Age 411 Ma is cited from Li et al. (2015)

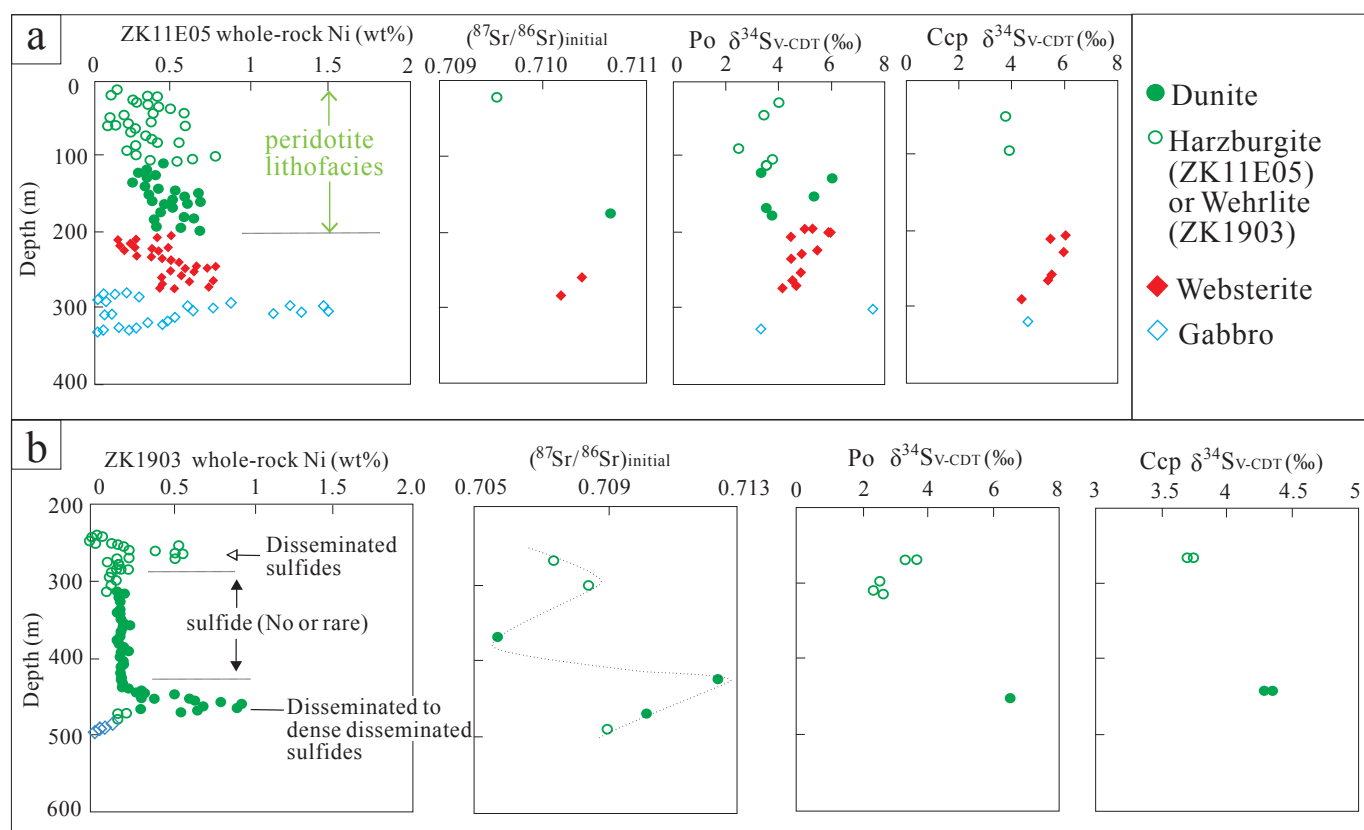


Fig. 10. The relationship between whole-rock Ni contents and in situ sulfur isotopes and whole-rock $(^{87}\text{Sr}/^{86}\text{Sr})_{\text{i}}$ in drill core ZK11E05 (a) and ZK1903 (b).

The $(^{87}\text{Sr}/^{86}\text{Sr})_{\text{i}}$ values of the dunite in drill core ZK1903 range from 0.70574 to 0.71295, with an average of 0.70974. The $(^{87}\text{Sr}/^{86}\text{Sr})_{\text{i}}$ values of the upper wehrlite are 0.70751 at 258 m and 0.70868 at 294 m (Fig. 10b), whereas the $(^{87}\text{Sr}/^{86}\text{Sr})_{\text{i}}$ value of the bottom wehrlite is 0.70920. In Figure 10b, the high Ni contents in drill core ZK1903 clearly correspond to high $(^{87}\text{Sr}/^{86}\text{Sr})_{\text{i}}$ values.

In situ S isotopes

The in situ isotopic values of sulfides from drill cores ZK11E05 and ZK1903 are listed in Table 3. The $\delta^{34}\text{S}_{\text{V-CDT}}$ values of the pyrrhotite in the harzburgite, dunite, websterite, and gabbro in drill core ZK11E05 are 2.5 to 4.0, 3.3 to 6.1, 4.5 to 5.9, and 3.3 to 7.7‰, respectively (Figs. 10a, 11). The mean $\delta^{34}\text{S}_{\text{V-CDT}}$ values of the pyrrhotite in the harzburgite, dunite, websterite, and gabbro are 3.5, 4.4, 5.0, and 5.5‰, respectively. Thus, the $\delta^{34}\text{S}_{\text{V-CDT}}$ values of the pyrrhotite increase gradually from the top to the bottom of the complex. The $\delta^{34}\text{S}_{\text{V-CDT}}$ values of the pyrrhotite in the websterite have a tendency to decrease with increasing depth (Fig. 10a).

The $\delta^{34}\text{S}_{\text{V-CDT}}$ values of the two chalcopyrites from harzburgite of drill core ZK11E05 are 3.8 and 3.8‰, with a mean value of 3.8‰ (Figs. 10a, 11). The $\delta^{34}\text{S}_{\text{V-CDT}}$ values of the

Whole-rock sulfur isotopes of Proterozoic rocks (marble, skarn, and schist) in the western part of East Kunlun

In-situ sulfur isotopes of Py in the Proterozoic granitic gneiss (Surrounding rock of Xiarihamu intrusion)

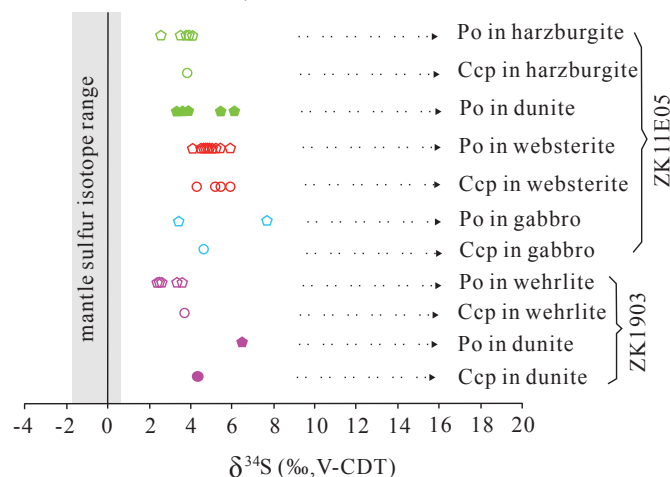


Fig. 11. Variations in $\delta^{34}\text{S}$ values of sulfides from the Xiarihamu deposit and typical Proterozoic rocks in the East Kunlun area. The whole-rock sulfur isotopes of Proterozoic rocks (marble, skarn, and schist) in the western part of East Kunlun are from Fang (2015). The $\delta^{34}\text{S}_{\text{V-CDT}}$ composition of the mantle is from Gao and Thiemens (1993) and Labidi et al. (2013). The other data are from the present study.

Table 3. In Situ S Isotopes

Drill core	Depth (m)	Rock type	Mineral	$\delta^{34}\text{S}_{\text{V-CDT}}(\text{‰})$	Mean value
ZK11E05	30	Harzburgite	Pyrrhotite	4.0	3.5
ZK11E05	45	Harzburgite	Pyrrhotite	3.5	
ZK11E05	89	Harzburgite	Pyrrhotite	2.5	
ZK11E05	105	Harzburgite	Pyrrhotite	3.7	
ZK11E05	108	Harzburgite	Pyrrhotite	3.6	
ZK11E05	120	Dunite	Pyrrhotite	3.3	4.4
ZK11E05	129	Dunite	Pyrrhotite	6.1	
ZK11E05	150	Dunite	Pyrrhotite	5.4	
ZK11E05	170	Dunite	Pyrrhotite	3.5	
ZK11E05	179	Dunite	Pyrrhotite	3.7	
ZK11E05	197	Websterite	Pyrrhotite	5.1	5.0
ZK11E05	197	Websterite	Pyrrhotite	5.3	
ZK11E05	201	Websterite	Pyrrhotite	5.9	
ZK11E05	201	Websterite	Pyrrhotite	5.9	
ZK11E05	207	Websterite	Pyrrhotite	4.5	
ZK11E05	223	Websterite	Pyrrhotite	5.5	
ZK11E05	228	Websterite	Pyrrhotite	4.9	
ZK11E05	234	Websterite	Pyrrhotite	4.5	
ZK11E05	254	Websterite	Pyrrhotite	4.8	
ZK11E05	263	Websterite	Pyrrhotite	4.5	
ZK11E05	270	Websterite	Pyrrhotite	4.7	
ZK11E05	274	Websterite	Pyrrhotite	4.1	
ZK11E05	301	Gabbro	Pyrrhotite	7.7	
ZK11E05	324.5	Gabbro	Pyrrhotite	3.3	
ZK11E05	45	Harzburgite	Chalcopyrite	3.8	3.8
ZK11E05	89	Harzburgite	Chalcopyrite	3.8	
ZK11E05	201	Websterite	Chalcopyrite	5.9	5.4
ZK11E05	207	Websterite	Chalcopyrite	5.4	
ZK11E05	223	Websterite	Chalcopyrite	5.9	
ZK11E05	254	Websterite	Chalcopyrite	5.4	
ZK11E05	259	Websterite	Chalcopyrite	5.3	
ZK11E05	283	Websterite	Chalcopyrite	4.4	
ZK11E05	316	Gabbro	Chalcopyrite	4.5	
ZK1903	266	Wehrlite	Pyrrhotite	3.6	2.9
ZK1903	266	Wehrlite	Pyrrhotite	3.3	
ZK1903	294	Wehrlite	Pyrrhotite	2.5	
ZK1903	305	Wehrlite	Pyrrhotite	2.4	
ZK1903	311	Wehrlite	Pyrrhotite	2.6	
ZK1903	457	Dunite	Pyrrhotite	6.4	6.4
ZK1903	266	Wehrlite	Chalcopyrite	3.7	3.7
ZK1903	266	Wehrlite	Chalcopyrite	3.7	
ZK1903	457	Dunite	Chalcopyrite	4.3	4.3
ZK1903	457	Dunite	Chalcopyrite	4.3	
The average sulfur isotope of sulfide from Xiarihamu mafic-ultramafic intrusion					4.5
ZK2105	216	Granitic gneiss	pyrite	11.2	11.2
ZK2105	216	Granitic gneiss	pyrite	11.2	
ZK2105	216	Granitic gneiss	pyrite	11.1	

chalcopyrite in the websterite range from 4.4 to 5.9‰, with an average of 5.4‰ (Figs. 10a, 11). The $\delta^{34}\text{S}_{\text{V-CDT}}$ values of the chalcopyrite in the websterite gradually decrease with increasing depth (Fig. 10a). The $\delta^{34}\text{S}_{\text{V-CDT}}$ value of one chalcopyrite sample from the gabbro is 4.6‰ (Fig. 11).

Due to the small size of the sulfide grains from drill core ZK1903 (<70 μm), most sulfides do not meet the requirements for an in situ sulfur isotope analysis. Thus, only rocks with sufficient sulfides were analyzed. The $\delta^{34}\text{S}_{\text{V-CDT}}$ values of the pyrrhotite in the wehrlite from drill core ZK1903 range

between 2.4 and 3.6‰, with a mean value of 2.9‰ (Fig. 10b). The $\delta^{34}\text{S}_{\text{V-CDT}}$ value corresponding to the high Ni wehrlite is significantly higher than that of the low Ni wehrlite (Fig. 10b). The $\delta^{34}\text{S}_{\text{V-CDT}}$ value of a pyrrhotite sample from the high Ni dunite is 6.4‰, and the $\delta^{34}\text{S}_{\text{V-CDT}}$ value of the chalcopyrite from the high Ni wehrlite range is 3.7‰ (Fig. 10b). Additionally, the $\delta^{34}\text{S}_{\text{V-CDT}}$ value of the chalcopyrite from the high Ni dunite is 4.3‰ (Fig. 10b).

In addition, this study also analyzed the in situ isotopic compositions of pyrite in the granitic gneiss located at 216 m

in drill core ZK2106, which yielded $\delta^{34}\text{S}_{\text{V-CDT}}$ values of 11.2, 11.2, and 11.1‰, with an average of 11.2‰ (Fig. 11).

Trace element analyses of zircon in the Xiarihamu eclogite

Zircon grains from the eclogite samples are subhedral and 100 to 200 μm in length. The internal structure of the selected metamorphic zircon crystals from the eclogites was studied with cathodoluminescence (CL) images. The grains exhibit nonconcentric zoning and are structureless. The LA-ICP-MS trace element compositions of zircon from the Xiarihamu eclogite are listed in Table 4. The U and Th contents of the zircons vary from 3.4 to 49.7 and 0.07 to 1.38 ppm, respectively, and the Th/U ratios are ~ 0.005 to 0.090. In a chondrite-normalized rare earth element (REE) diagram (Fig. 12), the zircons in the Xiarihamu eclogites exhibit positive Ce anomalies, have weak or nonexistent negative Eu anomalies, and are enriched in heavy REEs (HREEs) relative to light REEs (LREEs), with HREE/LREE values of 24.8 to 350.0.

Discussion

Geologic setting of the Xiarihamu mafic-ultramafic complex: Not an arc but a postcollisional setting

The geologic setting of the Xiarihamu deposit is in dispute. Some people think that it formed in an arc setting (Li et al. 2015), whereas other people hold the view that it formed in the postcollision environment (Wang et al., 2013; Wang, 2014; Peng et al., 2016; Song et al., 2016). We present new pieces of evidence to address this issue.

The Iridium subgroup of PGEs (IPGE; Ir, Ru, and Os) and Ni are more compatible in the mantle residue than the palladium subgroup (PPGE; Rh, Pt, and Pd) and Cu during partial melting of the mantle (e.g., Barnes et al., 1985). High degrees of partial melting would thus result in magmas with low Pd/Ir and high Ni/Cu ratios. In the plot of Pd/Ir versus

Ni/Cu, the Xiarihamu samples plot in the field of high Mg basalt (Fig. 13), indicating that the parental magmas of the Xiarihamu intrusion were most likely high Mg basalt in composition, in agreement with the predicted composition on the basis of olivine composition (Li et al., 2015 and “The influence of fractional crystallization of minerals on the sulfide saturation” on p. 1407).

However, based on the method of Li and Ripley (2011), Li et al. (2015) infer that the parental magma composition of Xiarihamu was as follows (all in wt %): $\text{SiO}_2 = 52.4$, $\text{TiO}_2 = 1.28$, $\text{Al}_2\text{O}_3 = 15.6$, $\text{FeO} = 6.54$, $\text{Fe}_2\text{O}_3 = 1.75$, $\text{MnO} = 0.64$, $\text{MgO} = 9.79$, $\text{CaO} = 8.91$, $\text{K}_2\text{O} = 0.90$, $\text{Na}_2\text{O} = 1.88$, $\text{P}_2\text{O}_5 = 0.30$, and $\text{NiO} = 0.05$. Li et al. (2015) interpreted this composition as boninitic. This may be because the major elements of boninite and high Mg basalt are similar. High Mg basalt is characterized by high MgO (>8%) and SiO_2 (51–55%) and low TiO_2 (usually <0.5%, some can reach 1.87%) and high field strength element contents (Francis et al., 1981; Sun et al., 1989; Cervantes and Wallace, 2003; Herzberg et al., 2007), whereas according to the classification of the International Union of Geological Sciences, boninite has the following features: (1) $\text{SiO}_2 > 52\%$, (2) $\text{MgO} > 8\%$, and $\text{TiO}_2 < 0.5\%$ (Le Bas, 2000). Therefore, we could not clearly distinguish the boninite and high Mg basalt by major elements. It is possible to distinguish these two types of rocks based on their geologic settings and trace elements (Sun et al., 1989; Srivastava, 2006). Moreover, the spinel in the boninite has some special features (Barnes and Roeder, 2001). In Figure 14, the spinel composition from the Xiarihamu complex is different from that of the boninites. Therefore, we infer that the parental magma of the Xiarihamu complex is not boninite but high Mg basaltic magma.

Li et al. (2015) thought that the Xiarihamu Ni deposit formed in an arc setting. The most important basis of this opinion is that the composition of the estimated parental magma is within the range of the arc boninite. However, the composition of spinel in the Xiarihamu complex is different from that of Alaskan-type complexes, which are produced in arc settings, and more similar to the spinel produced in a mid-ocean ridge

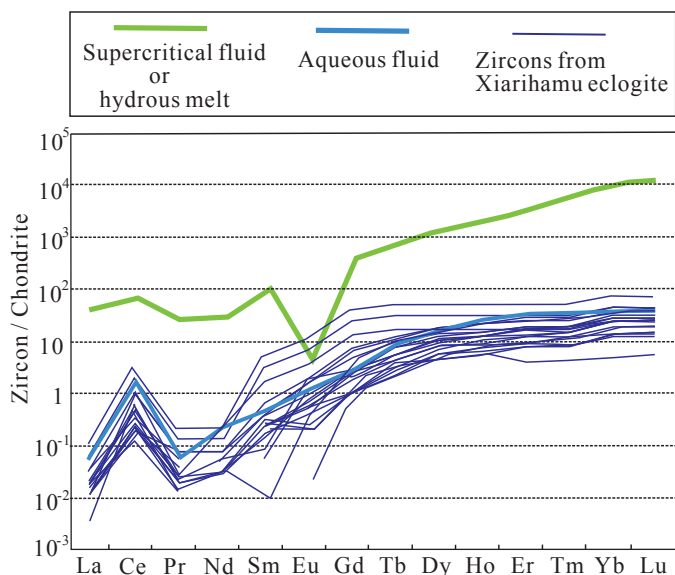


Fig. 12. Chondrite-normalized REE pattern for zircons from the Xiarihamu eclogite. Chondrite values after (McDonough and Sun, 1995). Supercritical fluid or hydrous melt data from the Dabie HP-UHP belt (Xia, 2009; Xia et al., 2010). Aqueous fluid data from the Sulu HP-UHP belt (Chen et al., 2011).

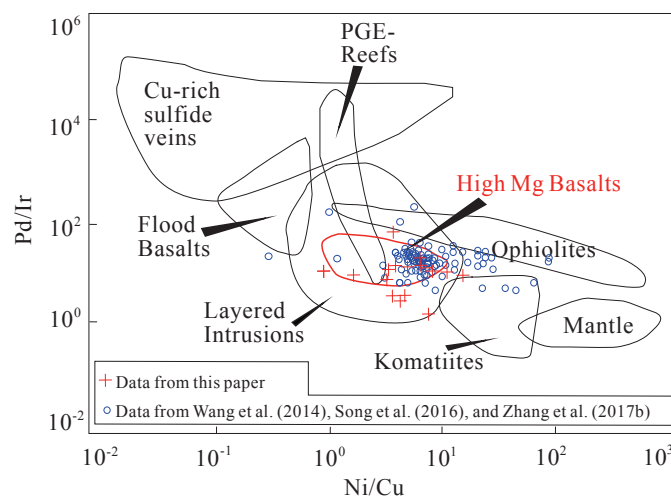


Fig. 13. Ni/Cu vs. Pd/Ir diagram of the Xiarihamu sulfides. Reference fields are after Barnes et al. (1985). Metal contents are corrected to 100% sulfide content after Barnes and Ripley (2016).

Table 4. LA-ICP-MS Trace Element Compositions of Zircon from Xiarhamu Eclogite

No.	La (ppm)	Ce (ppm)	Pr (ppm)	Nd (ppm)	Sm (ppm)	Eu (ppm)	Gd (ppm)	Tb (ppm)	Dy (ppm)	Ho (ppm)	Er (ppm)	Tm (ppm)	Yb (ppm)	Lu (ppm)	Th (ppm)	U (ppm)	Tb/U	HREE/LREE
E-1	0.008	0.047		0.033	0.057	0.079	1.17	0.54	5.68	1.30	3.98	0.57	4.96	0.95	0.2	14.0	0.015	91.1
E-2	0.003	0.060			0.007	0.065	0.45	0.27	2.68	0.82	2.46	0.43	3.63	0.67	0.1	15.2	0.008	87.4
E-3	0.008	0.026	0.009		0.019	0.030	0.74	0.36	4.07	1.03	3.25	0.45	4.09	0.79	0.1	4.4	0.017	169.7
E-4	0.007	0.044		0.015	0.045	0.023	0.13	0.12	1.86	0.53	1.84	0.31	2.52	0.43	0.2	3.4	0.057	58.4
E-5		0.043	0.003		0.073	0.079	1.15	0.42	4.32	1.29	3.94	0.77	6.43	1.26	0.1	6.3	0.008	105.1
E-6	0.001	0.248	0.012	0.056	0.180	0.193	2.37	0.70	6.99	1.96	6.95	1.15	9.98	1.70	0.9	35.4	0.027	49.4
E-7		0.047		0.015	0.053	0.019	0.48	0.28	4.49	1.62	5.76	0.96	6.62	1.01	0.2	6.7	0.030	162.4
E-8		0.049	0.005		0.032	0.012	0.27	0.12	1.22	0.33	1.21	0.23	2.37	0.58	0.1	8.8	0.015	67.7
E-9		0.052	0.006		0.070	0.059	0.96	0.34	3.37	1.13	3.24	0.59	5.36	1.03	0.1	12.7	0.012	90.3
E-10	0.003	0.029	0.002	0.014	0.067	0.070	0.93	0.48	3.59	1.11	3.36	0.65	5.42	0.92	0.2	6.7	0.025	93.6
E-11		0.068			0.018	0.045	0.65	0.22	2.30	0.62	1.68	0.34	2.75	0.52	0.1	12.7	0.005	74.5
E-12	0.047	0.073	0.005		0.037	0.028	0.54	0.25	2.95	0.80	2.91	0.47	3.65	0.71	0.2	11.4	0.021	67.4
E-13		0.170				0.064	0.79	0.38	4.53	1.29	4.60	0.82	5.91	0.91	0.7	9.9	0.073	85.7
E-14	0.022	0.435	0.005	0.140	0.736	0.642	7.67	1.87	12.64	2.68	7.66	1.22	9.91	1.47	1.4	49.6	0.028	26.7
E-15	0.014	0.088	0.011		0.072	0.033	0.52	0.23	2.72	0.80	2.22	0.38	3.44	0.64	0.2	6.7	0.029	52.8
E-16	0.007	0.187	0.002	0.023	0.013	0.121	1.54	0.54	4.77	1.48	4.71	0.78	7.42	1.19	0.6	21.8	0.027	67.9
E-17	0.003	0.049	0.002	0.014	0.048	0.012	0.36	0.26	2.95	0.82	2.91	0.44	3.95	0.65	0.4	7.3	0.058	98.5
E-18	0.016	0.174	0.002	0.087	0.130	0.114	0.94	0.27	3.55	0.79	2.46	0.44	3.04	0.59	1.0	22.8	0.044	24.8
E-19		0.042			0.019	0.047	0.18	0.13	1.06	0.30	0.65	0.11	0.88	0.15	0.4	4.1	0.090	33.7
E-20	0.004	0.044				0.001	0.21	0.19	2.78	1.01	4.42	0.74	6.50	1.18	0.2	4.1	0.040	350.0
E-21	0.010	0.119		0.014		0.096	0.86	0.34	4.18	1.28	4.12	0.57	5.22	0.94	0.6	12.4	0.050	77.0
E-22	0.004	0.059		0.015	0.001	0.032	0.93	0.48	4.21	1.16	3.77	0.80	5.99	1.15	0.3	8.6	0.032	174.8
E-23		0.303			0.123	0.145	1.98	0.68	6.19	1.78	5.59	0.99	7.77	1.54	0.4	25.2	0.017	49.9
E-24		0.069	0.002	0.035		0.018	0.25	0.16	2.10	0.67	2.21	0.35	2.79	0.49	0.4	5.8	0.075	74.5

Notes: Blank cells are below the detection limit

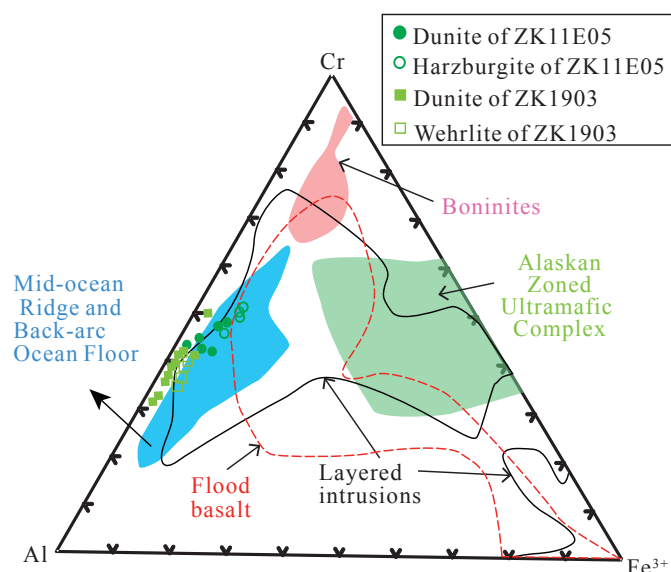


Fig. 14. Chromite trivalent ion plots for different kinds of rocks (Barnes and Roeder, 2001).

or back-arc ocean floor (Fig. 14). The $\text{Fe}^{3+}/\Sigma\text{Fe}$ ratio of the Xiarihamu spinel is <0.3 (Fig. 15); in contrast, the value for the Alaskan-type spinels is >0.3 . The higher $\text{Fe}^{3+}/\Sigma\text{Fe}$ ratio is consistent with a more oxidizing environment (Fudali, 1965; Brounce et al., 2014; Hartley et al., 2017). The $\text{Fe}^{3+}/\Sigma\text{Fe}$ ratio range of spinel from the Huangshandong Cu-Ni deposit, formed in a typical postcollisional setting, overlaps with that

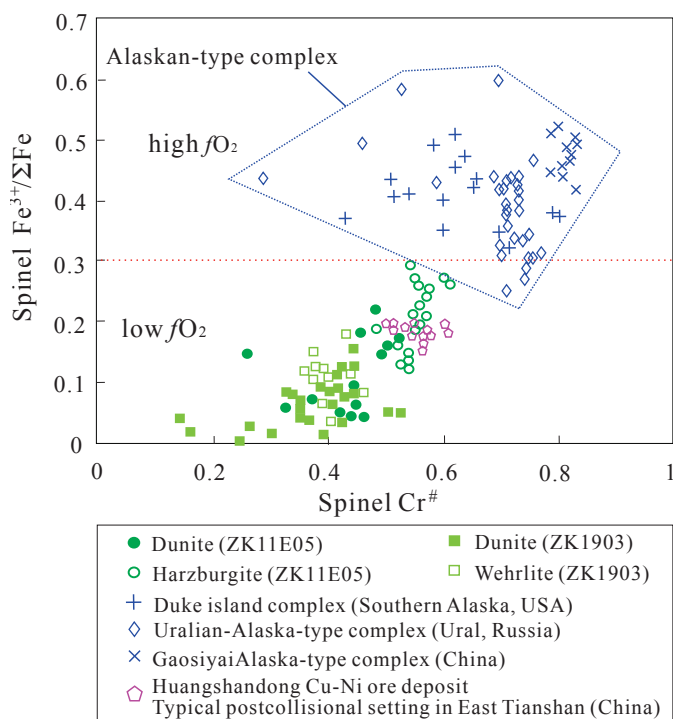


Fig. 15. Plot of spinel $\text{Fe}^{3+}/\Sigma\text{Fe}$ vs. spinel $\text{Cr}^\#$. Data compiled from the following sources: Duke Island complex (Li et al., 2013), Uralian-Alaskan-type complex in the Urals (Garuti et al., 2003), Huangshandong Cu-Ni ore deposit (Fu, 2012).

of Xiarihamu (Fig. 15). This may indicate that the Xiarihamu Cu-Ni deposit formed in a postcollisional setting.

The metamorphic peak age of the Wenquan eclogite in the East Kunlun orogenic belt is 428.0 ± 2.4 Ma (Meng et al., 2013), and the weighted average age of 16 points on the cores of zircons from the Xiarihamu eclogite is 436 Ma (Zhang et al., 2017a), which represents the high-pressure metamorphic peak age during continental collisions. The weighted mean age of 15 points on the margins of zircons from the Xiarihamu eclogite is 409 Ma, which represents the retrograde metamorphism age during the continental crust exhumation process (Zhang et al., 2017a). In the East Kunlun area, considerable mantle-related magmatic activity occurred after 428 Ma (Figs. 1b, 2). The products of this activity include Middle Silurian–Upper Silurian (428–419 Ma) basalt (Zhu et al., 2006; Qinghai Geological Survey, 2012), Lower Devonian–Middle Devonian (411.5–382.8 Ma) mafic dikes (Sun et al., 2004; Zhang et al., 2013; Xiong et al., 2014; Yang et al., 2014), and an Upper Silurian–Middle Devonian (419.0–391.1 Ma) A_2 -type granite (Chen et al., 2013; Liu et al., 2013b; Wang et al., 2013; Gan, 2014; Yan et al., 2016). The above mantle-related magmatic activity was interpreted to be a product of the postcollisional environment (Liu et al., 2012, 2013a; Gan, 2014; Peng et al., 2016; Song et al., 2016). Therefore, 428.0 Ma can be considered the transition from a continent-continent collisional setting to a postcollisional setting. Li et al. (2015) and Song et al. (2016) obtained ages of 411.6 ± 2.4 and 406.1 ± 2.7 Ma, respectively, for the Xiarihamu ore-bearing websterite. Therefore, it is reasonable that the Xiarihamu Ni deposit formed in a postcollisional setting.

The relationship between the Xiarihamu ultramafic intrusion and the eclogite

Li et al. (2015) and Song et al. (2016) obtained ages of 411.6 ± 2.4 and 406.1 ± 2.7 Ma, respectively, for the Xiarihamu ore-bearing websterite. Interestingly, eclogite with a peak eclogite facies metamorphic age of 436 Ma and a retrograde metamorphism age of 409 Ma was found near the Xiarihamu Ni sulfide deposit (Fig. 1c; Zhang et al., 2017a). Additionally, an eclogite with a metamorphic age of 428 Ma was also discovered in the eastern part of the East Kunlun orogenic belt (Meng et al., 2013). Based on the regional tectonic evolution, the peak eclogite facies metamorphic age 436 or 428 Ma suggests a deep, early Paleozoic subduction during the continent-continent collision, while the retrograde metamorphism age of 409 Ma suggests the age of the continental crust exhumation process in the East Kunlun orogenic belt (Raymond et al., 1989; Chopin, 2003; Meng et al., 2013; Qi et al., 2014; Zhang et al., 2017a). We believe the simultaneous appearance of the Xiarihamu magmatic Ni sulfide deposit and the eclogite exhumation in the same tectonic belt was not just a coincidence. Here, we assess this temporal and spatial relationship of the eclogite and the Xiarihamu Ni deposit by considering element transport and aqueous fluids and hydrous melt action during the subduction and exhumation of crustal rocks.

Crustal slabs are tectonically sheared in the subduction channel, causing metamorphic dehydration and partial melting, which produce aqueous fluids and hydrous melts (Zhao et al., 2007; Zheng, 2012). Aqueous fluids can only contain moderate amounts of LILEs, Sr, and Pb and cannot transport

significant amounts of trace elements, such as LREEs and Th. In contrast, hydrous melts can contain high contents of solutes, such as SiO_2 , Th, and LREEs (Hermann and Green, 2001; Hermann, 2002; Kessel et al., 2005; Hermann et al., 2006; Zhang et al., 2006; Klimm et al., 2008; Liu et al., 2009; Zheng et al., 2012). In the Sulu and Dabie HP-UHP belts, zircons that grew from aqueous fluids have flat HREE patterns, pronounced positive Ce anomalies and little or no Eu anomalies (Fig. 12), whereas zircons that grew from supercritical fluids (varying from hydrous melts to aqueous fluids) or were influenced by hydrous melts have steep HREE patterns, weak positive Ce anomalies, and obviously negative Eu anomalies (Xia, 2009; Xia et al., 2010; Chen et al., 2011; Fig. 12). The chondrite-normalized REE patterns of the zircons from the Xiarihamu eclogite are similar to the typical zircons that grew from aqueous fluids in the subduction channel.

Zircons influenced by hydrous melts in the Dabie HP-UHP rocks have high Th/U ratios of 0.14 to 0.85, whereas zircons influenced by aqueous fluids have low Th/U ratios of 0.02 to 0.08 (Xia et al., 2013) because Th has low solubility in aqueous fluids (Ayers et al., 1997; Kessel et al., 2005). The zircons in the northern Appalachian HP-UHP rocks equilibrated with a metamorphic aqueous fluid have a low Th/U ratio of <0.1 (Peterman et al., 2016), and the zircons in the Sulu HP-UHP rocks grown from aqueous fluids have an even lower Th/U ratio of <0.01 (W.C. Li et al., 2013). The Th/U ratios of zircons from the Xiarihamu eclogite range from 0.005 to 0.090, which are similar to those of zircons influenced by aqueous fluids.

Pearce et al. (2013) found that the trace elements and the deep dynamic processes differ considerably among arc, arc rift, and back-arc basin environments (Fig. 16). Figure 16a

shows the degree of addition of subducted material, and the fine lines paralleling the mantle element distribution lines represent the degree of addition of subduction zone components (including shallow aqueous fluid and deep hydrous melt). Figure 16b shows a plot that can be used to determine the degree of addition of components related to the shallow aqueous fluid in the subduction zone, and the contours parallel to the mantle array in the plot represent the additional percentage of shallow aqueous fluid. Figure 16a shows that the Xiarihamu ultramafic rocks include a significant subduction component. This subduction material could have been shallow aqueous fluids or deep hydrous melts. The addition of shallow aqueous fluids to the Xiarihamu ultramafic rocks is not supported by Figure 16b; thus, the Xiarihamu ultramafic rocks appear to have been influenced mainly by deep hydrous melts. Hydrous melts form mainly at high temperatures and ultrahigh pressures (>2 GPa), whereas aqueous fluids form at low temperatures and relatively low pressures (<2 GPa; Hermann and Green, 2001; Kessel et al., 2005; Zhang et al., 2006; Klimm et al., 2008; Liu et al., 2009; Zheng et al., 2012). The metamorphic pressure of the Xiarihamu eclogite was 2 GPa (Qi et al., 2014), which is consistent with the fact that the eclogite was influenced by shallow aqueous fluids. The Xiarihamu ultramafic rocks appear to have been influenced mainly by deep hydrous melts, meaning that the primary magma of the Xiarihamu ultramafic rocks was most likely derived from a deeper area with a pressure greater than 2 GPa.

According to modern plate tectonic theory, the formation of a typical collisional orogen via subduction is usually due to the subduction of oceanic crust beneath another plate, which promotes the production of island-arc magma in the

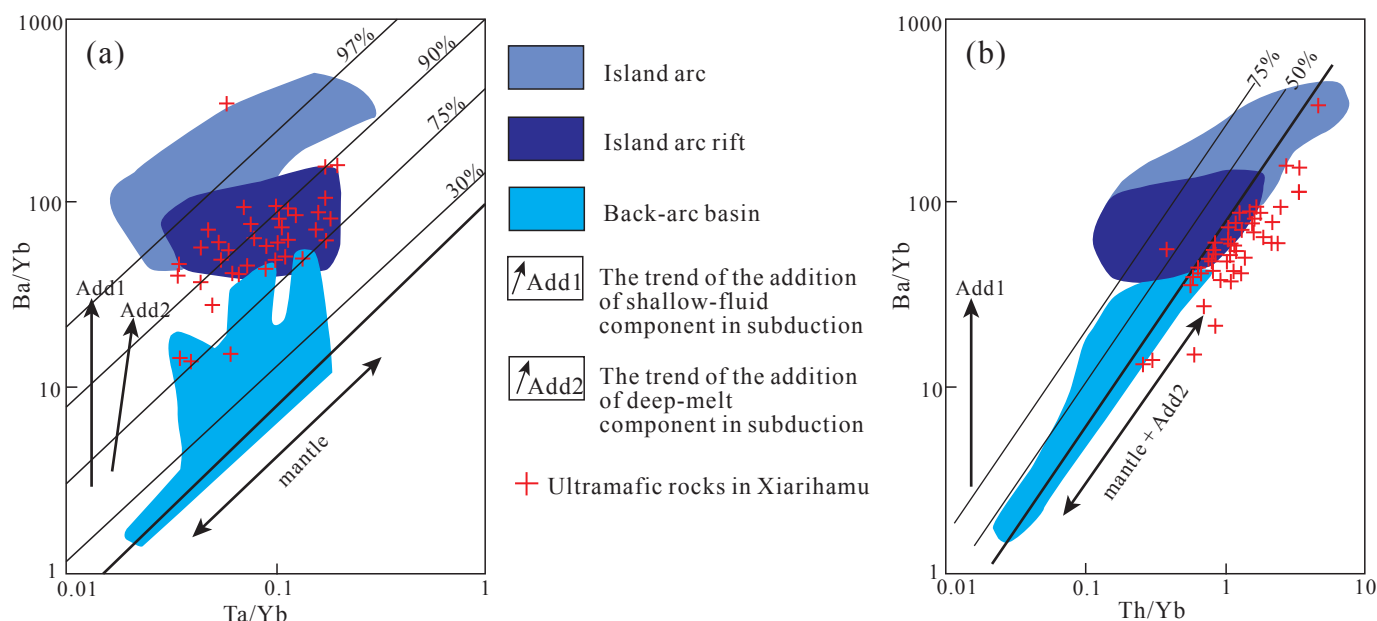


Fig. 16. Classification diagram for subduction material contamination (Pearce et al., 2013). a. Ba/Yb-Ta/Yb is a proxy for total subduction material addition (Pearce et al., 2013). b. Ba/Yb-Th/Yb is a proxy for the addition of the shallow aqueous fluid component during subduction (Pearce et al., 2013). Contours parallel to the mantle arrays in plots of M1/Yb vs. M2/Yb refer to the percentage of subduction-derived element M1 in the mantle source, assuming that M2 is subduction immobile ($=\Delta\text{M1}-\text{M2}$) (Pearce et al., 2013). In the Ba/Yb-Th/Yb plot, x% refers to the percentage of shallow subduction-derived Ba, assuming that the deep-melt subduction vector is subparallel to the MORB array (Pearce et al., 2013). The Xiarihamu ultramafic rock data are from (Wang, 2014; Jiang et al., 2015).

overriding plate. Additionally, continental crust attached to oceanic crust subducting beneath a continental plate will eventually collide with the continental crust of the overriding plate and can even be dragged deep into the subduction channel beneath the other plate. The oceanic crust and the subducted continental crust eventually break off at depth (Davies and Blanckenburg, 1995; Chemenda et al., 1996; Zhang et al., 2014). Although the exact exhumation mechanism of metamorphic rocks from the deep mantle to the shallow crustal levels is still in dispute, most researchers believe that without the break-off of oceanic and continental crusts, there is probably no uplifting and exhumation of continental crusts (Davies and Blanckenburg, 1995; Ernst, 2001; Xu et al., 2006; Zhang et al., 2008). After the continental crust completely detaches from the deeply subducted oceanic crust ("slab break-off"), as the density of the continental crust is lower than that of the surrounding rock in the upper mantle, the buoyancy effect can lead to its uplift (Davies and Blanckenburg, 1995; Ernst, 2001; Boutelier et al., 2004). Moreover, the extension or erosion of the upper crust and shortening of the lower crust are beneficial to the exhumation of the crust (Andersen and Jamveit, 1990; Balleve et al., 1990; Blake and Jayko, 1990; Ernst, 2001). Generally, slab break-off has been considered the most likely explanation for the exhumation of the HP-UHP rocks. Combined with all the evidence, this paper tries to use a slab break-off model to explain the simultaneous appearance of the Xiarihamu magmatic Ni sulfide deposit and the high-pressure eclogite exhumation in the East Kunlun orogenic belt as follows.

According to previous research on the tectonic evolution of the East Kunlun orogenic belt, the East Kunlun region entered the oceanic subduction stage after 508 Ma (Mo et al., 2007), and the transition from oceanic subduction to continent-continent collision is believed to have occurred at 438 Ma (Liu et al., 2013a). Thus, we infer that the period 508 to 438 Ma represents the oceanic crust subduction stage (Fig. 17a), and the trench-arc-basin system was developed during this period.

The Wenquan eclogite with a peak eclogite facies metamorphic age of 428 Ma and the Xiarihamu eclogite with a peak eclogite facies metamorphic age of 436 Ma (Meng et al., 2013; Zhang et al., 2017a) formed in the continent-continent collision stage (438–428 Ma; Fig. 17b). The subducting oceanic crust pulled the continental crust deep into the subduction channel, resulting in the formation of Xiarihamu eclogite (Fig. 17b). The Xiarihamu eclogite with a retrograde metamorphism age (exhumation age) of 409 Ma in East Kunlun (Zhang et al., 2017a) suggests that the break-off of oceanic and continental crusts probably occurred from 428 to 409 Ma, defining the initiation of postcollision (Fig. 17c).

The $\epsilon_{\text{Hf}(t)}$ values of the gabbro from the Xiarihamu II intrusion range from 6.0– to 2.9, with an average of 10.9 (Peng et al., 2016). Such high $\epsilon_{\text{Hf}(t)}$ values indicate that the Xiarihamu complex is unlikely to have been derived from the continental lithospheric mantle, which is characterized by $\epsilon_{\text{Hf}(t)}$ values of less than 0 (Griffin et al., 2000). Consequently, we conclude that the Xiarihamu mafic-ultramafic complex was probably derived from the depleted mantle or asthenospheric mantle. This is consistent with the view that the lithospheric mantle plays no active role in the formation of magmatic Ni-Cu

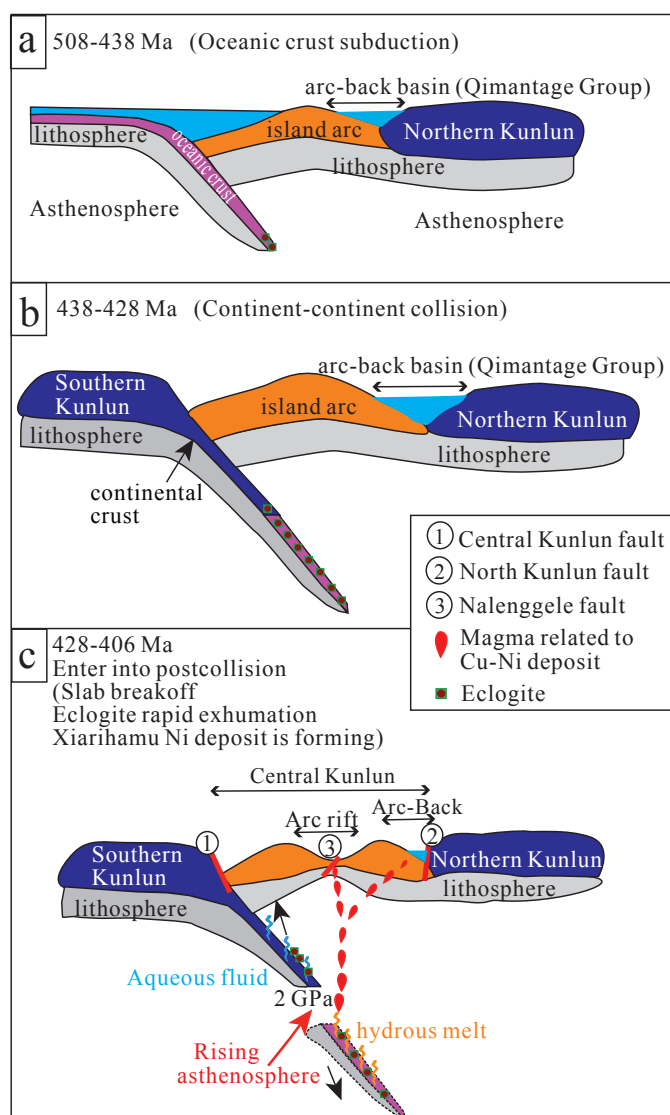


Fig. 17. The genetic connection between the Xiarihamu mafic-ultramafic complex and the eclogite.

deposits (Arndt, 2013). The Xiarihamu ultramafic rocks were influenced mainly by deep hydrous melts and derived from an area with a pressure higher than 2 GPa. In Figure 16, the Xiarihamu ultramafic rocks are shown to be located mainly in the island-arc rift or the back-arc field, and the spinel composition is located in the mid-ocean ridge or back-arc area (Fig. 14), suggesting that the Xiarihamu intrusion formed in an extensional environment. This is consistent with the view that the Xiarihamu Ni deposit formed in a postcollisional setting. Based on the all pieces of evidence, we infer that the following geologic process may have happened: the oceanic crust and the subducted continental crust broke off (Fig. 17c), and the continental crust carrying part of the eclogite (which was affected by aqueous fluids) began a rapid exhumation process. Meanwhile, the depleted mantle or asthenospheric mantle began to upwell. The primary magma of the Xiarihamu mafic-ultramafic complex formed from the depleted mantle or asthenospheric mantle influenced by hydrous melt

at a pressure higher than 2 GPa. Finally, the mafic-ultramafic intrusions formed in an arc rift or back-arc system in a post-collisional setting (Fig. 17c).

Sulfide segregation before complex emplacement

The Xiarihamu sulfide ores are characterized by high Ni tenor and extremely low PGE tenors ($\text{Ni}_{100\%}$ can reach up to 18.11%, $\text{PGE}_{100\%} = 46\text{--}235$ ppb; Table 1, Fig. 9). Song et al. (2016) inferred that the parental basaltic magma may have been PGE depleted when it was produced and that PGE-depleted basaltic magma with anomalously high Ni may have been produced by moderate-degree partial melting of the pyroxenite mantle in the orogenic belt. The olivine of the mantle peridotite is consumed by reaction with melts derived from the recycled oceanic crust to form a secondary pyroxenitic source (Sobolev et al., 2005). Magma originating from this source could have high Ni content (Sobolev et al., 2005). The interpretation that the magma was derived from a pyroxenitic mantle source could explain both the extremely low PGE tenors and high Ni/Cu ratios in the Xiarihamu deposit. However, an alternative explanation exists.

The sulfide liquid/silicate melt partition coefficients ($D^{\text{Sul/Sil}}$) of Cu ($1.06 \times 10^3\text{--}1.85 \times 10^3$) are much lower than those of Pd ($6.70 \times 10^4\text{--}5.36 \times 10^5$), which indicates that the Cu/Pd ratio will strongly increase once sulfide liquids segregate from the silicate magma (Mungall and Brenan, 2014). The Cu/Pd ratios of Xiarihamu ore rocks range from 1.2×10^5 to 2.6×10^5 ; these values are much higher than the primitive mantle Cu/Pd ratio (7,692) (McDonough and Sun, 1995), suggesting that sulfide segregation occurred in the early magma evolution.

Metal contents of sulfide liquids are a function of their abundances in parental silicate magmas, sulfide melt/silicate liquid partition coefficients, and the R factor (Campbell and Naldrett, 1979):

$$C_i^{\text{Sul}} = C_i^{\text{Sil}} \times D_i \times (R + 1)/(R + D_i), \quad (3)$$

where C_i^{Sul} and C_i^{Sil} represent the concentrations of element i in the sulfide melt and in the parental silicate magma, respectively; D_i is the sulfide melt/silicate liquid partition coefficient of element i ; and R is the ratio of the mass of silicate magma to the mass of sulfide that reached equilibrium together.

The Xiarihamu parental basaltic magma is assumed to be similar to high Mg basaltic composition, which has 14 ppb Pd (Hamlyn et al., 1985) and 300 ppm Ni (Francis et al., 1981; Zhao and Zhou, 2013). For modeling purpose, the partition coefficients of Pd and Ni between sulfide melt and silicate magma are 100,000 and 800, respectively (Peach et al., 1990; Mungall and Brenan, 2014). Palladium and Ni contents of the Xiarihamu sulfides are dramatically lower than the modeled values, plotting below the no sulfide removal dash line model 1 in Figure 18, indicating that the parental magma of Xiarihamu experienced early sulfide removal, consistent with the high Cu/Pd ratios of the sulfides. The metal contents in residual magmas after an earlier sulfide removal can be calculated on the basis of the Rayleigh Law: $(C_i^{\text{L}}/C_i^{\text{O}} = F^{(D_i - 1)})$, in which C_i^{O} and C_i^{L} represent abundances of element i in the original and residual magmas, respectively, and F is the mass fraction of segregated sulfide. Palladium and Ni contents in 100% sulfide liquid of the Xiarihamu sulfides from drill core

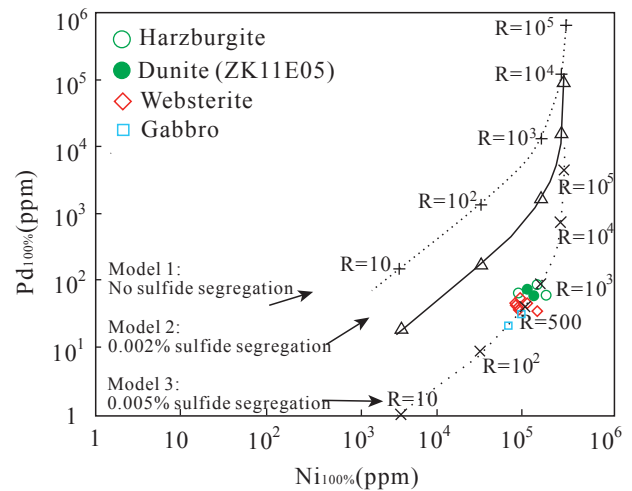


Fig. 18. Model calculations of the variation in $\text{Pd}_{100\%}$ with $\text{Ni}_{100\%}$ of the Xiarihamu sulfides from drill core ZK11E05. The Pd and Ni concentrations of the sulfides are corrected to 100% sulfide content after Barnes and Ripley (2016). Model 1 presents the sulfide melts segregated from PGE-rich basaltic magma with 14 ppb $\text{Pd}_{100\%}$, and 300 ppm $\text{Ni}_{100\%}$ under variable R values (labeled in the diagram). Model 2 displays sulfide liquids separated from PGE weakly depleted magma with 1.89 ppb $\text{Pd}_{100\%}$ and 295.2 ppm $\text{Ni}_{100\%}$ (because of a 0.002% sulfide removal from PGE-rich magma) under different R values. Model 3 shows the sulfide liquids produced by sulfide immiscibility in a heavily PGE depleted magma with 0.094 ppb $\text{Pd}_{100\%}$ and 288.3 ppm $\text{Ni}_{100\%}$ (because of a 0.005% sulfide removal from PGE-rich magma).

ZK11E05 plot well on the 0.005% earlier sulfide removal line with R factors between 200 and 1,000 (Fig. 18), indicating that the Xiarihamu parental magma may have achieved sulfide saturation and experienced approximately 0.005% sulfide segregation before complex emplacement.

To validate whether the above simulation is reasonable, we can verify the R value (the ratio of the mass of silicate magma to the mass of sulfide) based on the sulfur isotope signatures. The sulfur isotope values of sulfide in most magmatic Cu-Ni deposits are a mixture of mantle and crustal signatures. The relationship between the mixed sulfur isotope and R value can be expressed by equation (4) (Leshner and Burnham, 2001):

$$\delta^{34}\text{S}_{\text{V-CDT}}^{\text{mix}} = \frac{\delta^{34}\text{S}_{\text{V-CDT}}^{\text{mm}} \times C_{\text{mm}} \times \left(\frac{R}{1+R}\right) + \delta^{34}\text{S}_{\text{V-CDT}}^{\text{crust}} \times C_{\text{crust}} \times \left(\frac{1}{1+R}\right)}{C_{\text{mm}} \times \left(\frac{R}{1+R}\right) + C_{\text{crust}} \times \left(\frac{1}{1+R}\right)}, \quad (4)$$

where $\delta^{34}\text{S}_{\text{V-CDT}}^{\text{mix}}$ is the mixed sulfur isotope ratio; $\delta^{34}\text{S}_{\text{V-CDT}}^{\text{mm}}$ and $\delta^{34}\text{S}_{\text{V-CDT}}^{\text{crust}}$ are the initial sulfur isotope ratios of the parental magma derived from the mantle and the crust, respectively; and C_{mm} and C_{crust} are the initial sulfur abundances in the parental magma derived from the mantle and the crust, respectively.

As one of the end members chosen in this study, the pyrite (FeS_2) in the country rock (granitic gneiss) has a $\delta^{34}\text{S}_{\text{V-CDT}}$ value of 11.2‰ (from this paper) and an S content of 54 wt % {54 wt % = 2° (atomic weights of sulfur)/[atomic weights of iron + 2° (atomic weights of sulfur)]}. The DM, the other end member, has a $\delta^{34}\text{S}_{\text{V-CDT}}$ value of -1.8‰ (Labidi et al., 2013), and the basaltic magma derived from mantle usually has an S concentration of 1,000 ppm (Keays, 1995; Leshner and Stone, 1996). Hence, in equation (4), the values are as follows:

$\delta^{34}\text{S}_{\text{V-CDT}}^{\text{min}} = -1.8\text{‰}$, $C_{\text{min}} = 1,000$ ppm, $\delta^{34}\text{S}_{\text{V-CDT}}^{\text{crust}} = 11.2\text{‰}$, and $C_{\text{crust}} = 0.54\%$. The R is between 200 and 1,000 and averages nearly 500 (Fig. 18). For R = 200, 1,000, and 500, the $\delta^{34}\text{S}_{\text{V-CDT}}^{\text{mix}} = 7.66$, 2.74, and 4.93‰, respectively, calculated by equation (4). The in situ $\delta^{34}\text{S}_{\text{V-CDT}}$ value of sulfides from the Xiarihamu I complex ranges from 2.4 to 7.7‰ with an average of 4.5‰. Therefore, the sulfur isotopes calculated from the R values are nearly the same as those from the in situ sulfur isotope test. Thus, our simulation is reasonable.

Sulfide segregation before complex emplacement can explain the extremely low PGE tenors. Actually, sulfide segregation before complex emplacement can also explain the high Ni/Cu ratios. The sulfide liquid/silicate melt partition coefficient ($D^{\text{Sul/Sil}}$) for Cu (1.06×10^3 – 1.85×10^3) (Mungall and Brenan, 2014) is higher than that for Ni (575–836) (Peach et al., 1990). Thus, the segregation of a sulfide phase will deplete the magma in PGE and Cu more than in Ni (Leshner and Stone, 1996), causing a high Ni/Cu ratio. In summary, the Xiarihamu parental magma may have achieved sulfide saturation and experienced approximately 0.005% sulfide segregation before complex emplacement or may have been derived from a pyroxenite mantle.

SiO₂ and CaCO₃ contamination

At depths of 361 m in drill core ZK15E05 and 119 m in drill core ZK13E05, in the eastern part of the cross section in Figure 4, granitic gneiss xenoliths are observed in the dunite (Fig. 4e–f), and the pyroxene and sulfides occur at the contact between the olivine and the granitic gneiss. Furthermore, at a depth of 443 m in drill core ZK1903, in the western part in the cross section in Figure 4, we observe small calcite (marble) xenoliths wrapped in the clinopyroxene in the dunite (Fig. 5a). Therefore, there is no doubt that granitic gneiss and marble contamination occurred in the eastern and western parts, respectively.

The average SiO₂ and CaO contents of the dunite olivine in drill core ZK11E05 are 40.52 and 0.03%, respectively, whereas the average SiO₂ and CaO contents of the dunite olivine in drill core ZK1903 are 40.05 and 0.04%, respectively. The relationship between the olivine CaO content and the coexisting melt CaO content, in wt %, follows the following rule (Libourel, 1999):

$$\text{CaO}_{\text{Olivine}} = 0.0877 * (e^{(0.106 * \text{CaO}_{\text{melt}})} - 1). \quad (5)$$

That is, the more Ca rich the melt, the more Ca rich the olivine (Libourel, 1999). The olivine in dunite from drill core ZK11E05 has a higher SiO₂ content and a lower CaO content relative to that from drill core ZK1903. Therefore, it is reasonable to infer that the magma in drill core ZK1903, in the western part of the cross section, was contaminated by more marble. This conclusion is consistent with the fact that the western part of the complex is underlain by marble and small calcite (marble) xenoliths enclosed in the clinopyroxene in the dunite (Fig. 5a). This contamination has some influence on the lithofacies distribution. The pyroxene in the dunite from drill core ZK1903 is clinopyroxene ($\text{CaMgSi}_2\text{O}_6$), which is enriched in Ca (Fig. 5a), and wehrlite is mainly distributed in ZK1903. Thus, we infer that the following reaction may have occurred: $(\text{Mg, Fe})\text{SiO}_4(\text{melt}) + 2\text{CaCO}_3 + 3\text{SiO}_2 = \text{CaMgSi}_2\text{O}_6(\text{Cpx}) + \text{CaFeSi}_2\text{O}_6(\text{Cpx}) + 2\text{CO}_2$. Therefore, the

addition of marble promoted the crystallization of clinopyroxene in the dunite. In contrast, the pyroxene in the dunite and harzburgite from drill core ZK 11E05 is orthopyroxene ($\text{Mg}_2\text{Si}_2\text{O}_6$) (Fig. 5e). The SiO₂ content of the granitic gneiss is approximately 73.5% (Gan, 2014; Wang et al., 2016); thus, we infer that the following reaction may have occurred: $\text{Mg}_2\text{SiO}_4(\text{melt}) + \text{SiO}_2 = \text{Mg}_2\text{Si}_2\text{O}_6(\text{Opx})$. In this way, the granitic gneiss contamination promoted the crystallization of orthopyroxene in the dunite. Therefore, contamination by different crustal components probably plays an important role in the formation of different lithofacies and olivines with different compositions.

Sulfide saturation was triggered or promoted by SiO₂ addition in the Kalatongke Cu–Ni ore deposit, Noril'sk Cu–Ni ore deposit, and Biamazhai Cu–Ni deposit (Lightfoot and Hawkesworth, 1997; Zhang et al., 2003; Wang et al., 2005). Sulfides occur at the contact between the olivine and the granitic gneiss xenolith (Fig. 4f); therefore, contamination by SiO₂ is also likely beneficial to sulfide saturation in Xiarihamu. The $(^{87}\text{Sr}/^{86}\text{Sr})_i$ ratio ($t = 411$ Ma) of surrounding rocks of the Xiarihamu intrusion is 0.778818, calculated from the $(^{87}\text{Sr}/^{86}\text{Sr})_i$ ratio of Neoproterozoic rocks ($t = 942$ Ma; Chen et al., 2007), whereas the $(^{87}\text{Sr}/^{86}\text{Sr})_i$ ratio of the depleted mantle is 0.7027 (Salters and Stracke, 2004), which is much lower than that of surrounding rocks. Thus, the intrusion with a higher crustal contamination degree will show a higher $(^{87}\text{Sr}/^{86}\text{Sr})_i$ ratio. The $(^{87}\text{Sr}/^{86}\text{Sr})_i$ ratio of the dunite with a high Ni content is higher than that of the harzburgite with a low nickel content in drill core ZK11E05 (Fig. 10a). The area with an abrupt increase in the Ni content in drill core ZK1903 is associated with an obvious increase in the $(^{87}\text{Sr}/^{86}\text{Sr})_i$ ratio (Fig. 10b). The high Ni content of the websterite in drill core ZK11E05 also corresponds to a high $(^{87}\text{Sr}/^{86}\text{Sr})_i$ ratio (Fig. 10a). Consequently, crustal contamination is beneficial to mineralization process.

Crustal sulfur contamination

Crustal sulfur contamination is crucial in most magmatic Cu–Ni ore deposits (Leshner et al., 1984; Leshner and Stone, 1996; Leshner and Burnham, 2001; Holwell et al., 2007; Leshner and Barnes, 2008; Keays and Lightfoot, 2010; Fiorentini et al., 2012; Ripley and Li, 2017). The $\delta^{34}\text{S}_{\text{V-CDT}}$ value of the DM is -1.80‰ (Labidi et al., 2013), and the average $\delta^{34}\text{S}_{\text{V-CDT}}$ values of enstatite and ordinary and carbonaceous chondrites are -0.26 ± 0.07 , -0.02 ± 0.06 , and $+0.49 \pm 0.16\text{‰}$ (Gao and Thiemens, 1993). Thus, a range of -1.80‰ to $+0.49\text{‰}$ roughly represents the $\delta^{34}\text{S}_{\text{V-CDT}}$ composition of the mantle (Fig. 11). In contrast, the in situ $\delta^{34}\text{S}_{\text{V-CDT}}$ value of pyrite in the granitic gneiss from drill core ZK2105 (216 m) is 11.2‰ (Table 3). The in situ $\delta^{34}\text{S}_{\text{V-CDT}}$ value of sulfides from the Xiarihamu I complex range from 2.4 to 7.7‰, with an average of 4.5‰, indicating that the Xiarihamu I complex experienced considerable crustal sulfur contamination. The mean $\delta^{34}\text{S}_{\text{V-CDT}}$ values of the pyrrhotite in the harzburgite, dunite, websterite, and gabbro from drill core ZK1105 are 3.45, 4.41, 4.98, and 5.5‰, respectively. The increase in the average sulfur isotope values among the different rock types is accompanied by increases in the Ni and S contents (Fig. 10b). Consequently, crustal sulfur contamination plays an important role in the mineralization.

A range of $\delta^{34}\text{S}_{\text{V-CDT}}$ mixing proportions can be computed with the aid of a simple mixing equation (Ripley et al., 1999):

$$\delta^{34}\text{S}_{\text{V-CDT}}^{\text{mix}} = \frac{C_{\text{crust}} \times D}{C_{\text{crust}} \times D + C_{\text{mm}} \times (1-D)} \times \delta^{34}\text{S}_{\text{V-CDT}}^{\text{crust}} + \frac{C_{\text{mm}} \times (1-D)}{C_{\text{crust}} \times D + C_{\text{mm}} \times (1-D)} \times \delta^{34}\text{S}_{\text{V-CDT}}^{\text{mm}}, \quad (6)$$

where C_{crust} and C_{mm} refer to the sulfur concentrations of the crust and the parent magma derived from the mantle, respectively; $\delta^{34}\text{S}_{\text{V-CDT}}^{\text{crust}}$ and $\delta^{34}\text{S}_{\text{V-CDT}}^{\text{mm}}$ are the $\delta^{34}\text{S}_{\text{V-CDT}}$ values of the crust and the parent magma derived from the mantle, respectively; and D is the degree of crustal sulfur contamination.

As one of the end members chosen in this study, the basaltic magma derived from the mantle has an S concentration of 1,000 ppm (Keays, 1995; Leshner and Stone, 1996) and a $\delta^{34}\text{S}_{\text{V-CDT}}$ value of 0.49‰ (Gao and Thiemens, 1993). The crust, the other end member, has an S concentration of 404 ppm (the average value of continental crust from Rudnick and Gao (2003)). The Proterozoic strata in the western segment of the East Kunlun orogenic belt have the highest whole-rock $\delta^{34}\text{S}_{\text{V-CDT}}$ value record, with a range of 10.1 to 18.3‰ (Fig. 11), as shown by the whole-rock sulfur isotope analysis of marble, skarn, and schist (Fang, 2015). Thus, the highest whole-rock $\delta^{34}\text{S}_{\text{V-CDT}}$ value found in the East Kunlun orogenic belt is 18.3‰. In the Xiarihamu area, the pyrite in the country rock (granitic gneiss) has a $\delta^{34}\text{S}_{\text{V-CDT}}$ value of 11.2‰, and the average $\delta^{34}\text{S}_{\text{V-CDT}}$ value of the sulfides in the Xiarihamu mafic-ultramafic intrusion is 4.5‰, which is between the $\delta^{34}\text{S}_{\text{V-CDT}}$ value of the mantle and the crust (Fig. 11). Thus, during the simulation, if we choose a larger $\delta^{34}\text{S}_{\text{V-CDT}}$ value for the mantle and a larger $\delta^{34}\text{S}_{\text{V-CDT}}$ value for the crust, a lower crustal sulfur contamination degree will be obtained. Therefore, we can determine the minimum crustal sulfur contamination degree with the following two end member parameters: (1) we selected +0.49‰ as the $\delta^{34}\text{S}_{\text{V-CDT}}$ value of the basaltic magma derived from mantle instead of -0.18‰; (2) the maximum sulfur isotope value of (18.3‰) in the East Kunlun orogenic belt is selected as the $\delta^{34}\text{S}_{\text{V-CDT}}$ value of the crust end member.

Hence, in the above equation, the values are as follows: $\delta^{34}\text{S}_{\text{V-CDT}}^{\text{crust}} = 18.3‰$, $C_{\text{crust}} = 404$ ppm, $\delta^{34}\text{S}_{\text{V-CDT}}^{\text{mm}} = 0.49‰$, and $C_{\text{mm}} = 1,000$ ppm. In this way, the minimum crustal sulfur contamination degree is 41.8%. If the $\delta^{34}\text{S}_{\text{V-CDT}}^{\text{crust}} = 11.2‰$ (the $\delta^{34}\text{S}_{\text{V-CDT}}$ of the pyrite in the granitic gneiss in the Xiarihamu area), $C_{\text{crust}} = 404$ ppm, $\delta^{34}\text{S}_{\text{V-CDT}}^{\text{mm}} = 0.49‰$, and $C_{\text{mm}} = 1,000$ ppm, the crustal sulfur contamination degree is 59.7%. Therefore, we infer that crustal sulfur most likely accounts for 40 to 60% of the total sulfur content on average in the Xiarihamu deposit.

We need to emphasize that the crustal contamination degree is not equal to the crustal sulfur contamination degree. Using the two end member Hf isotope mixing model, Li et al. (2015) revealed that up to 15% crustal contamination is permitted for the parental magma of the Xiarihamu ultramafic rocks. This crustal contamination degree is much lower than the minimum crustal sulfur contamination degree (41.8%) calculated in the present paper. This difference is probably due to the parental magma having been contaminated by some rocks enriched in sulfur before emplacement.

Fractional crystallization history of the Xiarihamu complex

The Fo values of the olivine in the peridotite in two drill cores decrease from the central zone to the upper and lower marginal zones (Figs. 6a, 7a, c, 8a). However, we cannot arbitrarily conclude that the olivine in the central zone crystallized first because the waning and waxing flow rates and temperature of a magma may have caused the later olivine to exhibit a high Fo value (Leshner, 2007). For example, at the Raglan Ni-Cu-(PGE) deposit associated with komatiitic rocks, the host unit crystallized from the bottom upward; this is supported by the fact that where Fo increases upward, the olivine is reversely zoned, whereas where Fo decreases upward, olivine is normally zoned (Leshner, 2007). This difference probably exists because the Fo content is being controlled by variations in flow rate: when the flow rate increases, then Mg content in the magma increases and the Fo increases, whereas the reverse occurs when flow rate decreases (Leshner, 2007). The olivines in the lower zone of the peridotite (including dunite, harzburgite, and wehrlite) of Xiarihamu have normal zoning (Fig. 7a, c), suggesting that the olivines have crystallized downward from the central zone of the peridotite to the lower zone of the peridotite.

Normal zoning was found at the core of the olivine in the upper marginal and central zones of peridotite, whereas the rim of the olivine in the central and upper marginal zones shows clear and weak reverse zoning, respectively (Fig. 7a, c). If the olivine was transported or crystallized at depth, the Fo value in the core would be stable, as observed in the olivine cores of kimberlite (Fedortchouk and Canil, 2004; Bussweiler et al., 2015) or ultramafic lamprophyres (Jaques and Foley, 2018; Nosov et al., 2018). The olivine cores of kimberlite are considered to be xenocrysts in the kimberlite magma, whereas the margins represent cognate olivine crystallized from the kimberlite melt itself (Kamenetsky et al., 2008; Pilbeam et al., 2013). However, no xenocryst feature was observed, and the Fo value decreased or increased gradually in the core or rim in the Xiarihamu olivine, respectively (Fig. 7a, c). Therefore, the Xiarihamu olivine probably crystallized in situ.

The Fo values of the olivine in the peridotite in two drill cores decrease from the central to the upper marginal zone (Fig. 7a, c); moreover, the Fo values of olivine core have normal zoning in both the central and the upper marginal zones and are higher than the Fo values of the olivine rim in the upper marginal zone (Fig. 7a, c). Therefore, on the whole, the olivine crystallized upward from the central to the upper marginal zone. The presence of minerals with normal and reverse zoning in a single sample reveals two magma mixing processes (Stronck et al., 2009) or the waning and waxing flow rates and temperature of the magma (Leshner, 2007). If the two ultramafic magmas mixed, it is most likely that the olivine compositions would have been altered and that the Fo values would be highly variable in the area of mixing. However, the curve of the olivine Fo values in drill core ZK1903 is smooth (Fig. 7c). Therefore, it is more likely that the reverse zoning of the olivine rim was caused by an increased magma flow rate and temperature. Magma with a higher flow rate and temperature usually has a higher Mg content (Leshner, 2007). The reverse zoning is more obvious in the central peridotite, and the Fo in the olivine rim was even higher than that in the olivine core

in the central peridotite drill core ZK1903 (Fig. 7a, c). Therefore, during the late period of crystallization, olivine most likely crystallized downward from the upper marginal to the central peridotite.

Because the Mg number values of the orthopyroxene in the websterite in drill core ZK11E05 tend to decrease downward (Figs. 6c, 7b) and the orthopyroxene exhibits normal zoning (Fig. 7b), orthopyroxene crystallized downward.

The influence of fractional crystallization of minerals on the sulfide saturation

The occurrence of sulfides in silicate melts requires sulfide saturation in the magma system. The sulfur content in the melt when sulfide saturation was achieved is called SCSS. Based on high-temperature and pressure experiments, Liu et al. (2007) presented a model (eq. 7) for the prediction of the SCSS in the melt:

$$\ln(S \text{ in ppm})_{\text{SCSS}} = 11.35251 - \frac{4454.6}{T} - 0.03190 \frac{P}{T} + 0.71006 \ln(\text{MFM}) - 1.98063[(\text{MFM})(X_{\text{H}_2\text{O}}^{\text{melt}})] + 0.2186 \ln(X_{\text{H}_2\text{O}}^{\text{melt}}) + 0.36192 \ln(X_{\text{FeO}}^{\text{melt}}), \quad (7)$$

where P is in the bar, T is in K, and MFM is a compositional parameter describing the melt based upon the following cation mole fractions:

$$\text{MFM} = \frac{\text{Na} + \text{K} + 2(\text{Ca} + \text{Mg} + \text{Fe}^{2+})}{\text{Si} \times (\text{Al} + \text{Fe}^{3+})},$$

$X_{\text{H}_2\text{O}}^{\text{melt}}$ is the mole fraction of water in the melt and $X_{\text{FeO}}^{\text{melt}}$ is the mole fraction of FeO in the melt.

Li and Ripley (2009) presented an alternative empirical equation for the SCSS based on the available experimental results. The SCSS conforms to equation (8):

$$\ln X_S = -1.76 - 0.474 (10^4/T) - 0.021P + 5.559X_{\text{FeO}} + 2.565XTiO_2 + 2.709XCaO - 3.192XSIO_2 + 3.049XH_2O, \quad (8)$$

where P is in kbars, T is in degrees Kelvin, X is the mole fraction, and Xi is the mole fraction of the content of oxide i in the melt.

Because fractional crystallization influences the magma composition, the fractional crystallization process can also affect the sulfide saturation (Scoates and Mitchell, 2000). Previous research suggests that the consumption of FeO during the fractional crystallization of olivine and spinel is an important factor in sulfide saturation (Park et al., 2004; Wykes et al., 2015; Liu et al., 2017). The regions with rapidly decreasing olivine Fo (Fig. 8a) and clinopyroxene Mg number values (250–300 and 450–490 m; Fig. 8c) in drill core ZK1903 correspond well with the locations of orebodies, implying that the crystallization of olivine or clinopyroxene played a role in the sulfide saturation. In drill core ZK11E05, the whole-rock Ni content of the peridotite is inversely related to the Cr number and FeO contents in spinel, and the whole-rock Ni content of the websterite is negatively correlated with the orthopyroxene Mg number value; thus, during the crystallization of spinel and orthopyroxene, more sulfide formed. The presence of sulfides in spinel and orthopyroxene (see Fig. 5c-f) also strongly supports this hypothesis.

Using a partition coefficient of 11 for Ni between olivine and basaltic magma, which is nearly the average value of

experimental and observed results for basaltic systems (Pedersen, 1979; Nabelek, 1980), our modeling shows that the olivine Fo-Ni relationship could not be explained by the fractional crystallization of olivine alone (Fig. 19). Using a partition coefficient of 800 for Ni between sulfide melt and basaltic magma, which is within the range results from Peach et al. (1990) and Patten et al. (2013), our modeling results show that most Fo-Ni values fall in the area with mass olivine/sulfide ratios of 10 to 50 (Fig. 19). The trend of the Fo-Ni relationship is most consistent with the curve with a mass olivine/sulfide ratio of 25 (Fig. 19). Moreover, Li et al. (2015) found that a primary sulfide inclusion was enclosed within olivine crystal in drill core ZK902; thus, the Xiarihamu magma was saturated with immiscible sulfide liquid when the primitive olivine was crystallizing from the magma.

The distribution coefficient between sulfides and silicates ($D_{\text{Sul/Sil}}$) ranges from 4×10^5 to $2-3 \times 10^6$ for platinum group elements (PGEs) (Mungall and Brenan, 2014). Patten et al. (2013) suggested that the $D_{\text{Sul/Sil}}$ of Ni is 776. Therefore, Ni and PGEs are highly compatible in sulfides, and the earliest sulfide occurring in the melt would have the highest PGE and Ni contents. In drill core ZK11E05, the peridotite with the highest olivine Fo value also has the highest sulfide PGE_{100%} and Ni_{100%} values (Fig. 9). Therefore, we believe that the Xiarihamu magma was sulfide saturated when the olivine began to crystallize. We do not claim that the initial sulfide saturation was caused by the fractional crystallization during the Xiarihamu complex emplacement and consolidation process, but we believe that the fractional crystallization promoted a higher fraction of sulfide formation. To determine the amount of sulfur precipitated from the melt during crystallization differentiation, we performed the following simulation.

During fractional crystallization, both the change in the melt composition and the decrease in temperature could affect the SCSS. Which is the greater factor? To solve this problem, we used the MELTS program to evaluate the effect of the change in melt composition and temperature on the

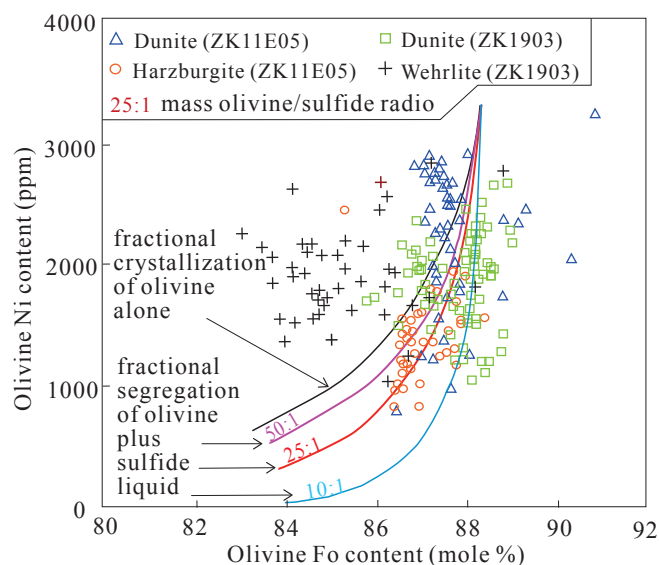


Fig. 19. Quantitative simulation of olivine fractional crystallization and concurrent sulfide segregation with different mass olivine/sulfide ratios.

SCSS during crystallization. The parental magma's composition was required for this purpose. Based on the method of Li and Ripley (2011), Li et al. (2015) inferred the parental magma composition of the Xiarihamu, but the composition does not contain Cr_2O_3 ; therefore, it could not reflect the spinel crystallization. Moreover, the parental magma composition inferred by Li et al. (2015) does not contain H_2O . The H_2O content in olivine-hosted melt inclusions trapped at upper to middle crustal pressures (1–6 kbars) from a subduction-related high Mg basalt in central Mexico is 1.3 to 5.2 wt % (Cervantes and Wallace, 2003). Therefore, it is reasonable that the parental magma contains some water. These reasons have prompted us to make an independent estimation of the parental magma composition for the Xiarihamu complex. In our calculations, we first excluded samples containing >0.25 wt % Ni and >0 wt % loss-on-ignition (LOI) to reduce errors associated with sulfide correction and the effect of severe alteration as suggested by Li and Ripley (2011). The $\text{FeO}/\text{FeO}_{(\text{total})}$ value is assumed to be 0.9 in the magma. The most primitive olivine (i.e., Fo = 88.1) was selected from a harzburgite sample at 70 m in drill core ZK11E05. This rare sulfide feature and the low Ni content (0.23%) of this sample can prevent the Fe-Ni exchange between olivine and sulfide liquids. We used the method of Chai and Naldrett (1992), the composition of the most primitive olivine (i.e., Fo = 88.12), the olivine liquid Fe-Mg exchange coefficient $[\text{Kd}, (\text{FeO}/\text{MgO})_{\text{Ol}}/(\text{FeO}/\text{MgO})_{\text{Liq}}]$ of 0.3 ± 0.03 by Roeder and Emslie (1970), and the average composition of the selected samples

from Wang (2014), Jiang et al. (2015), Li et al. (2015) and found that the MgO and FeO contents in the parental magma were 8.58 and 6.79 wt %, respectively. We could draw a best-fit line through the relationship between each major element and MgO based on the major element data for the selected samples (Chai and Naldrett, 1992). Then, we could calculate the original content of each major element based on a primary MgO content of 8.58 wt %. Thus, we could determine the liquid composition of the parental magma (all in wt %): MgO = 8.58, FeO = 6.79, Fe_2O_3 = 0.75, SiO_2 = 52.62, Al_2O_3 = 15.06, CaO = 10.48, K_2O = 0.39, MnO = 0.19, Na_2O = 0.28, Cr_2O_3 = 0.11, P_2O_5 = 0.14, TiO_2 = 0.62, and H_2O = 4.00.

We have simulated the crystallization sequence of our parental magma for the Xiarihamu intrusion between 1,300° and 1,050°C, using the rhyolite-MELTS program (<http://melts.ofm-research.org/macossx.html>) (Ghiorso and Sack, 1995; Gualda et al., 2012; Ghiorso and Gualda, 2015) under the assumed conditions of 1 kb and FMQ + 1 oxidation state (Fig. 20). The SCSS is calculated by equations (7) and (8), respectively. Olivine, spinel, orthopyroxene, and clinopyroxene occur at 1,174°, 1,150°, 1,102°, and 1094°C, respectively (Fig. 20). When orthopyroxene appears, olivine stops crystallizing. The SCSS decreases during crystallization (Fig. 20); therefore, if sulfide saturation was achieved when olivine began to crystallize, sulfide would have segregated with olivine. This is consistent with our view that the fractional crystallization promoted more sulfide formation. During the fractional crystallization process, two factors are changing:

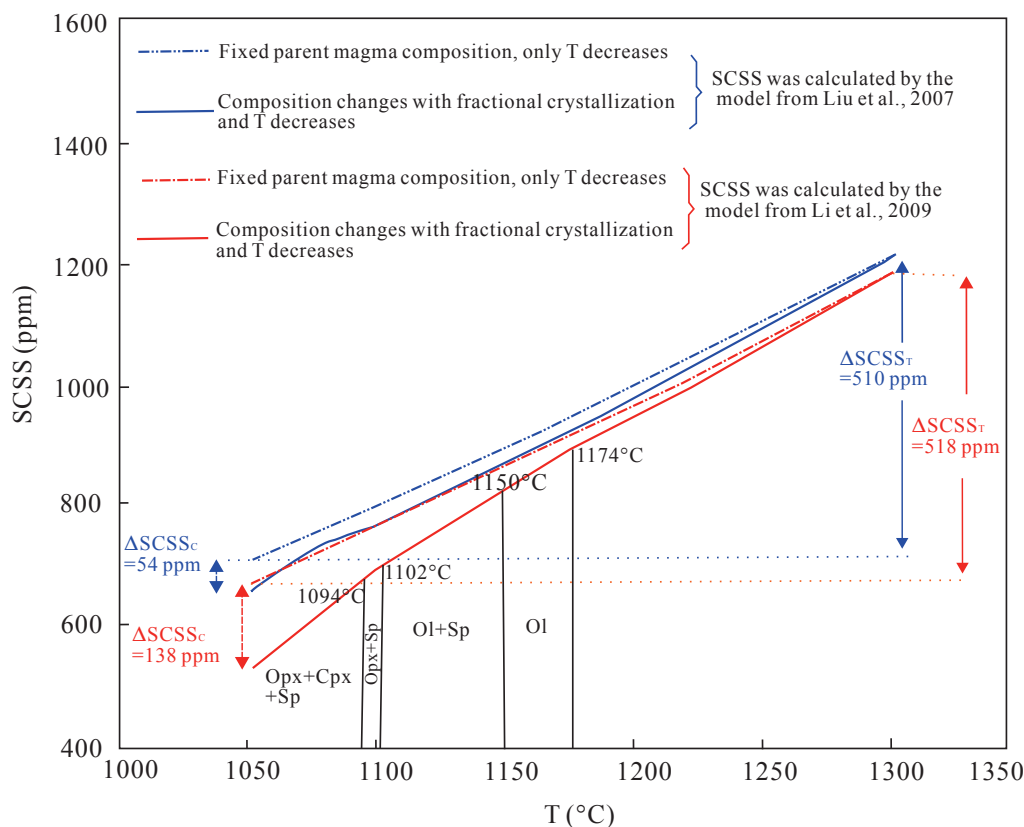


Fig. 20. Model of the variation in the sulfur contents at sulfide saturation (SCSS) during the fractional crystallization process. The modeling details were provided in the text. Cpx = clinopyroxene, Ol = olivine, Opx = orthopyroxene, Sp = spinel.

temperature and melt composition. To assess the effects of these two factors on sulfide saturation, we fixed the parental magma's composition and only let the temperature decrease. At each temperature, there was a corresponding SCSS value, named $SCSS_T$. When the temperature decreases from T_1 to T_2 , the temperature's influence on SCSS could be estimated by the $\Delta SCSS_T = SCSS_{T_1} - SCSS_{T_2}$. The SCSS value during the normal fractional crystallization (not a fixed parental magma composition), named $SCSS_{normal}$, minus the $SCSS_T$, will be the contribution of the composition change to the SCSS, we named it as $\Delta SCSS_C = SCSS_{normal} - SCSS_T$. In Figure 20, when the temperature decreases from 1,300° to 1,050°C, the $\Delta SCSS_T$ values calculated by equations (7) and (8) are 510 and 518 ppm, respectively, while the $\Delta SCSS_C$ values calculated by equations (7) and (8) are 54 and 138 ppm, respectively. Thus, the contribution of the temperature on the SCSS during fractional crystallization is $510/(510 + 54) = 90\%$ or $518/(518 + 138) = 79\%$. Therefore, the decrease in SCSS during fractional crystallization depends mainly on the decrease in temperature, and the change in melt composition is a secondary or minor factor in the decrease in SCSS.

From 1,300° to 1,050°C, the $SCSS_{normal}$ ($\Delta SCSS_C + SCSS_T$) value calculated by equations (7) and (8) decreases by 564 and 656 ppm, respectively. Thus, approximately 565 to 656 ppm sulfur is precipitated from melt and becomes a constituent of sulfide. The average S content of the Xiarihamu ultramafic rocks is approximately 2% (Table 1). Therefore, the contribution of fractional crystallization to sulfide saturation may be approximately $656 \text{ ppm}/2\% = 3.3\%$. At 1,300°C, the SCSS is approximately 1,200 ppm (Fig. 20). Even if all the S could be precipitated from the melt, the contribution of fractional crystallization to sulfide saturation would be approximately $1,200 \text{ ppm}/2\% = 6.0\%$. We have inferred that crustal sulfur most likely accounts for 40 to 60% of the total sulfur content on average in the Xiarihamu deposit. Thus, the contribution of fractional crystallization to sulfide saturation is small compared to crustal sulfur contamination.

If the olivine experiences gravity settling, a high Fo value would have been observed at the bottom of the peridotite. However, the Fo value at the bottom is much lower than the central zone (Fig. 8a). If the sulfide experienced gravity settling, then there would be no sulfides at the top of the lithofacies, such as the sulfide in the upper marginal peridotite in drill core ZK1903 (Fig. 8a). Therefore, neither olivine nor sulfide showed gravity-settling features. Although the fractional crystallization promoted only a small portion of sulfide formation, the rich orebody with the most enriched sulfide was located where the latest crystallization occurred in a lithofacies. This result may have occurred because sulfur is an incompatible element; thus, increasingly more sulfides were accumulated at the latest crystallization.

Sulfide liquid fractionation and SO_2 degassing

Sulfide liquid fractionation has been observed in some giant Cu-Ni deposits, such as Noril'sk and Sudbury (Ballhaus et al., 2001; Mungall, 2007; Kalugin and Latypov, 2012). In most cases, the first mineral to crystallize from a cooling sulfide melt will be monosulfide solid solution (mss), and the residual sulfide liquid is enriched in Cu (Holwell and McDonald, 2010; Naldrett, 2011).

The distribution coefficient between mss and sulfide liquid ($D_{mss/Sul}$) ranges from 0.07 to 0.12 for Pd (Mungall et al., 2005) and from 3.2 to 12.4 for Ir (Mungall et al., 2005; Liu and Brennan, 2015). Therefore, it is reasonable that the sample with a larger mss fraction will have a lower Pd/Ir value. The $Mg^\#$ values of the orthopyroxenite (Fig. 6c) and Pd/Ir values of sulfides (Fig. 9) tend to decrease downward in the websterite in drill core ZK11E05. Thus, we infer that, during the orthopyroxene fractional crystallization, sulfide liquid fractionation occurred, and more mss formed at the late stage (Fig. 21).

The $\delta^{34}S_{V-CDT}$ values of the pyrrhotite and the chalcopyrite in the websterite in drill core ZK11E05 decrease downward (Fig. 10a). However, the S contents of the websterite increase downward (Fig. 9), and the $\delta^{34}S_{V-CDT}$ value of pyrite in the country rock is 11.2‰, which is much higher than the $\delta^{34}S_{V-CDT}$ values of sulfides in the Xiarihamu mafic-ultramafic rocks. Therefore, the decrease in the $\delta^{34}S_{V-CDT}$ values cannot be explained by a decrease in the amount of the crustal sulfur. From equation (4), we can see that the R factor has an effect on the sulfur isotope composition of sulfides. However, the sulfide contents of the websterite increase downward, meaning the R factor decreases downward. When the R factor decreases, the $\delta^{34}S_{V-CDT}$ will increase. This effect is in stark contrast to the $\delta^{34}S_{V-CDT}$ variation trend of sulfides in websterite. Therefore, the influence of R factor can be ruled out. Sulfides formed in situ by magmatic segregation from an overlying silicate melt commonly show a ^{32}S enrichment (low $\delta^{34}S_{V-CDT}$ values) in the early-formed sulfides (Seccombe et al., 1981). However, under a lower oxygen fugacity condition, magmatic fractionation should yield a slight ^{34}S enrichment in early-formed sulfides (Ohmoto and Rye, 1979). This trend has been obtained in the Scotia nickel deposit, Western Australia (Seccombe et al., 1981). Li (2016) detected that the highest H_2S content is present in the websterite among different rock

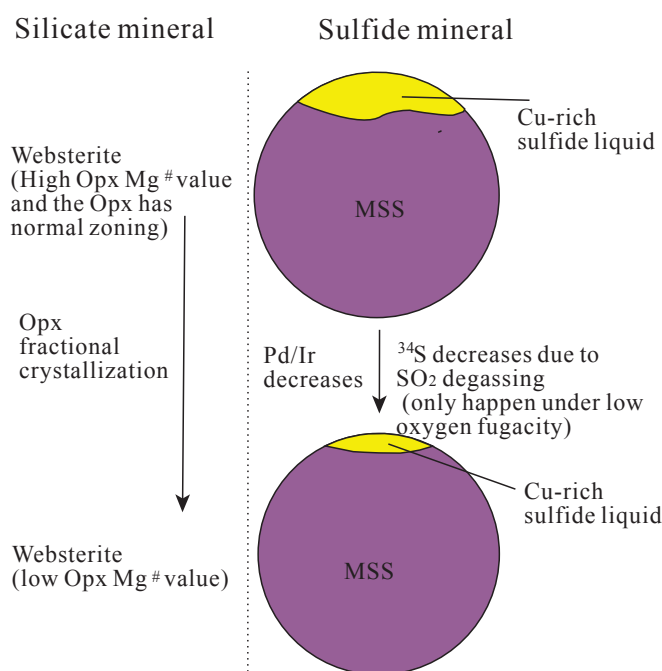


Fig. 21. Sulfide liquid fractionation in the websterite.

types in the Xiarihamu deposit. Therefore, the websterite probably crystallized under low oxygen fugacity.

Some researchers have noted that $\delta^{34}\text{S}_{\text{V-CDT}}$ can be affected by magma degassing (Torssander, 1989; Kyser, 1990; Li et al., 2009). A melt dominated by S^{2-} has lower ^{34}S than SO_2 gas if isotopic equilibrium is attained (de Moor, 2013). Therefore, if the emitted gas is primarily SO_2 , the residual sulfide $\delta^{34}\text{S}_{\text{V-CDT}}$ value will decrease (Yallup et al., 2013; Piochi et al., 2015). This probably happened under low oxygen fugacity because loss of a vapor containing $\text{SO}_2\text{--H}_2\text{S}$ in any proportion for a SO_4^{2-} -dominated magma (high oxygen fugacity) will lead to a ^{34}S enrichment of the melt in either an open or closed system (Mandeville et al., 2009). Fractionation between SO_2 and low amounts of sulfide sulfur in the melt is the only possible fractionation that could lead to a ^{34}S depletion of the melt (Mandeville et al., 2009). The fractionation factors (where x and y are different S species) can be approximated by the following relationships (T is in K) (Taylor, 1986):

$$\delta^{34}\text{S}_{\text{SO}_2}-\delta^{34}\text{S}_{\text{H}_2\text{S}} = 1000 \ln \alpha_{\text{SO}_2-\text{H}_2\text{S}} = -0.42(10^3/T)^3 + 4.367(10^3/T)^2 - 0.105(10^3/T) - 0.41, \quad (9)$$

and

$$\delta^{34}\text{S}_{\text{S}^{2-}}-\delta^{34}\text{S}_{\text{H}_2\text{S}} = 1000 \ln \alpha_{\text{S}^{2-}-\text{H}_2\text{S}} = -1.1(10^3/T)^2 + 0.19 \quad (10)$$

Equation 9 holds in the interval 400° to 1300°C , whereas equation 10 is valid in the range of 600° to $1,000^\circ\text{C}$ (Taylor, 1986). The combination of the equations yields:

$$1000 \ln \alpha_{\text{SO}_2-\text{S}^{2-}} = 1000 \ln \alpha_{\text{SO}_2-\text{H}_2\text{S}} - 1000 \ln \alpha_{\text{S}^{2-}-\text{H}_2\text{S}}, \quad (11)$$

and the fractionation factors between the emitted SO_2 gas and the S^{2-} in the melt can be approximated by the following equation (T is in K):

$$\delta^{34}\text{S}_{\text{SO}_2}-\delta^{34}\text{S}_{\text{S}^{2-}} = 1000 \ln \alpha_{\text{SO}_2-\text{S}^{2-}} = -0.42(10^3/T)^3 + 5.467(10^3/T)^2 - 0.105(10^3/T) - 0.6, \quad (12)$$

Thus, at $1,000^\circ\text{C}$, the $\delta^{34}\text{S}_{\text{SO}_2}-\delta^{34}\text{S}_{\text{S}^{2-}} = 2.49\text{‰}$. This process will decrease the ^{34}S content in the melt. Therefore, we infer that the $\delta^{34}\text{S}_{\text{V-CDT}}$ values decreasing downward in the websterite may be the result of SO_2 degassing from the melt under low oxygen fugacity (Fig. 21). The loss of isotopically heavy sulfur through magmatic degassing of SO_2 -rich volatiles has also been found in Archean komatiites (Caruso et al., 2017).

Dynamic sulfide saturation process during complex emplacement

We propose a dynamic sulfide saturation process during the Xiarihamu complex emplacement. The details of the process and a basis for our views are as follows.

The PGE-depleted magma intruded into the Neoproterozoic rocks (granitic gneiss and marble; Fig. 22a). The magma experienced approximately 0.005% sulfide segregation and was contaminated by sulfur-rich rocks before emplacement. The earliest olivine crystal occurred at a nearly central location within the peridotite. The magma in the eastern part was contaminated with granitic gneiss (Fig. 22a). That is, SiO_2 was the component that the rocks in drill core ZK11E05 assimilated. In contrast, the magma in the western part was contaminated with both granitic gneiss and marble (Fig. 22a), thus incorporating both SiO_2 and CaCO_3 . The $\delta^{34}\text{S}_{\text{V-CDT}}$ value of the country rock is 11.16‰, and crustal sulfur

was added to the magma system. Crustal sulfur accounts for approximately 40 to 60% of the sulfur content of the whole-rock system. Contamination occurred in all of the stages and did not end until the complex was totally consolidated. The Xiarihamu magma was saturated with immiscible sulfide liquid when the primitive olivine was crystallizing from the magma. Therefore, in our model, sulfide liquid also existed at the place where the earliest olivine formed (Fig. 22a). The Fo values of the olivine in the peridotite in two drill cores decrease from the central zone to the lower marginal zones, and the olivine grains have normal zoning (Fig. 7a, c), indicating that the olivine crystallized from the central to the lower marginal zones (Fig. 22a-c). On the whole, the olivine crystallized upward from the central to the upper marginal zone (Fig. 22b-c) because the Fo values of olivine in the peridotite in two drill cores decrease from the central to the upper marginal zone and the Fo values of olivine core are normally zoned in the both the central and the upper marginal zones of peridotite (Fig. 7a, c). Reverse zoning at the olivine rim in the upper and the central zones of peridotite and the Fo in the olivine rim is even more prevalent than that in the olivine core in the central zone of peridotite in drill core ZK1903, which may indicate that the rock crystallized downward from the upper marginal to the central zone during the late period of olivine crystallization (Fig. 22c).

The spinel occurs mainly along the edges of olivine grains (Fig. 5f-1, f-2) or as an interstitial material between olivine grains (Fig. 5e). Therefore, olivine crystallized first and the spinel crystallized later. Because the spinel Cr number and FeO contents in drill core ZK11E05 decrease downward (Fig. 6b), the spinel most likely crystallized earlier at the top and later at the bottom (Fig. 22b-c). The spinels did not experience gravity settling. Although the spinel Cr number values in drill core ZK1903 do not show an obvious decrease downward (Fig. 8b), the FeO contents decrease gradually from top to bottom (Fig. 8b), as observed in drill core ZK11E05 (Fig. 6b). Thus, the spinel in drill core ZK1903 also likely crystallized downward. Most of the spinel Cr number values from drill core ZK1903 are lower than that of harzburgite in drill core ZK11E05, and the wehrlite in ZK1903 has the lowest spinel Cr number value (Fig. 15). Therefore, the earliest spinel crystallization occurred in the harzburgite (ZK11E05; Fig. 22b). The orthopyroxene Mg number value in the peridotite of drill core ZK11E05 does not show any clear trend (Fig. 6c), implying that the orthopyroxene crystallized randomly (Fig. 22b-c). The clinopyroxene Mg number value in the peridotite of drill core ZK1903 decreases both downward and upward (Fig. 8c), indicating that the clinopyroxene crystallized progressively from the central to the marginal zone (Fig. 22c). During the mineral fractional crystallization, the temperature was decreasing and the melt composition was changing. These two factors, mainly the decrease in temperature, have led to a decrease in SCSS in the magma system. Although the fractional crystallization only promoted a small part of sulfide formation, the orebody with the most enriched sulfide was located where the latest crystallization occurred in the peridotite (Fig. 22b-c). This pattern may exist because sulfur is an incompatible element.

When the peridotite was nearly consolidated, the websterite intruded into the peridotite, supported by crosscutting

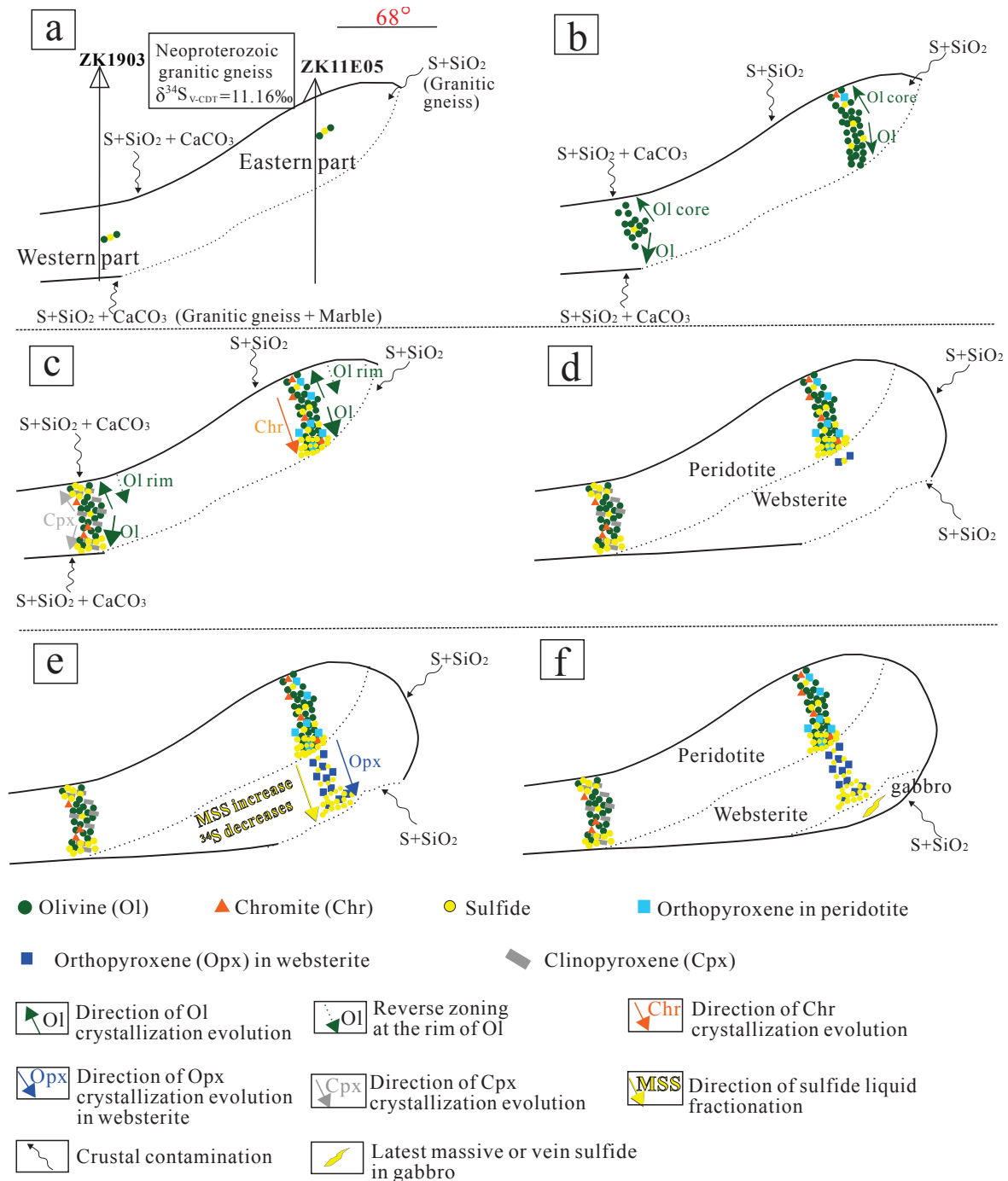


Fig. 22. Consolidation and sulfide saturation process of the Xiarihamu I mafic-ultramafic complex.

relationships (Fig. 4h-i). Because the $\text{Mg}^\#$ values of the coarse-grained orthopyroxene in the websterite in drill core ZK11E05 tends to decrease downward (Fig. 6c) and the orthopyroxene exhibits normal zoning (Fig. 7b), the orthopyroxene crystallized earlier at the top (Fig. 22d). The whole-rock Ni content is negatively correlated with the orthopyroxene $\text{Mg}^\#$ value (Fig. 6c) and positively correlated with the S content (Fig. 9), indicating that more sulfide reached saturation or sulfide gathered during the orthopyroxene crystallization (Fig. 22d-e). During the orthopyroxene fractional crystallization,

the sulfide liquid fractionation most likely also occurred, and more mss formed at the late stage; meanwhile, SO_2 degassing may have happened under low oxygen fugacity, resulting in the $\delta^{34}\text{S}$ decrease; thus, the sulfides formed at the late stage have a low $\delta^{34}\text{S}_{\text{V-CDT}}$ value.

When the websterite was completely consolidated, the gabbro intruded into the websterite, supported by the crosscutting relationship (Fig. 4j). There are fragments of angular gabbros where the gabbro is in contact with the massive orebodies (Fig. 4k), and the massive sulfide cuts through the plagioclase (Fig.

5k-1, k-2); thus, some massive ore intruded into the gabbro when the gabbro was completely consolidated (Fig. 22f).

Conclusions

Important conclusions from this study are listed below:

1. The Xiarihamu Cu-Ni deposit formed in a postcollisional setting.
2. The slab break-off model could explain the simultaneous appearance of the Xiarihamu magmatic Ni sulfide deposit and the eclogite exhumation in the East Kunlun orogenic belt.
3. The Xiarihamu parental magma may have achieved sulfide saturation and experienced approximately 0.005% sulfide segregation at depth or may have been derived from a pyroxenite mantle.
4. The crustal sulfur most likely accounts for approximately 40 to 60% of the total sulfur content in the Xiarihamu magma.
5. The decrease in SCSS during fractional crystallization mainly depends on the decrease in temperature, and the change in melt composition is a secondary or minor factor during fractional crystallization in the decrease in SCSS.
6. The contribution of fractional crystallization to sulfide saturation or total sulfur may be approximately 3.3 to 6.0%. Although the fractional crystallization promoted only a small portion of sulfide formation, increasing amounts of sulfides were accumulated during the later crystallization because sulfur is an incompatible element. This is most likely the reason why the rich orebody with the most enriched sulfide was located where the latest crystallization occurred in a lithofacies.

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